

RESEARCH ARTICLE OPEN ACCESS

The Impact of a Buyer's Incentives and Organizational Structure on Suppliers' Quality

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Received: 17 May 2023 | **Revised:** 20 July 2025 | **Accepted:** 10 August 2025

Handling Editor: Manpreet Hora

Keywords: knowledge | organizational structure | quality improvement | supply chain

ABSTRACT

Changes in a buyer organization's incentives and structure can have a significant impact on supply chain performance, such as the quality of sourced components. We examine this relationship using data from a unique quasi-experiment where one of the two plants of a buyer introduced long-term quality improvement incentives and a more organic structure for supply chain engineers, while the other plant did not. Analyzing longitudinal component quality data from suppliers to both plants using difference-in-differences and triple-differences methods, we find that (i) the changes in incentives and structure led to differential improvements in supplier quality across the two plants, and (ii) suppliers did not easily transfer quality knowledge gained from one buyer relationship to others. Our paper contributes by showing that organizational redesign by a buyer focused on implicit incentives along with a supportive organic organizational structure can create a learning environment at its suppliers that can help improve the quality of sourced components and the learning generated does not necessarily spill over to other buyers.

1 | Introduction

The quality of sourced components is important to every aspect of a firm's subsequent value chain, including the quality of the final product. Extant literature has suggested many ways to achieve improved supplier quality, such as contracting mechanisms and better relationships with suppliers (Balachandran and Radhakrishnan 2005). However, the effect of organizational incentives coupled with changes in the organizational structure of a buyer on supplier quality improvement has received limited attention in the literature. In this paper, we investigate the supplier-side quality response to buyer-side changes in organizational structure and incentives, both of which can significantly impact the actions and performance of line engineers and managers, by examining a real-life longitudinal experiment of Auto Inc., a multinational automotive manufacturer (the firm requested anonymity). We used data on quality (defects) of

sourced components from a unique quasi-experiment where this buyer with two plants instituted changes at one while keeping the organizational structure and incentives of the other unchanged. Our investigation suggests that enhanced quality control in sourcing (and consequently lower defect levels) can be achieved through implicit incentives focused on long-term improvement and developing organizational structures dedicated to empowering decision-making for exploratory actions to create better quality knowledge between a firm and its suppliers (Throughout the manuscript we use the “vendor” and “supplier” interchangeably).

Two key elements known to influence organizational actions and employee performance are incentives (Gambardella et al. 2015; Lee and Meyer-Doyle 2017; Kretschmer and Puranam 2008) and organizational structure (Csaszar 2013; Zhang et al. 2012; Pleshko and Nickerson 2008; Marengo and Pasquali 2012).

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Both financial incentives (e.g., bonuses, stock options) and non-financial incentives (e.g., recognition, career paths, performance goals) shape employee behavior by aligning effort with desired outcomes (De Leeuw and Van Den Berg 2011; da Silveira et al. 2013; Steinbach et al. 2017; Cadsby et al. 2007). This is particularly relevant in supply chain contexts, where long-term quality-focused incentives can encourage buyer engineers to engage in sustained root-cause-driven collaboration with suppliers. Such actions contribute to the development of quality knowledge in suppliers, ultimately improving the quality of the components. Although prior research has explored the general effects of incentives on performance, the effect of implicit incentives at a buyer on the performance of suppliers has been less studied. In particular, the role of such implicit incentives in enabling deeper learning and quality improvement at suppliers remains underexplored. Our study addresses this gap by examining how such implicit incentives affect supplier quality performance in the context of organizational redesign.

Prior research in organizational theory has also examined how organizational structure helps shape internal decision making (Burton et al. 2006), and sourcing literature has explored the dynamics of buyer-supplier relationships (Helper and Sako 1995). However, the effect of a firm's organizational structure on suppliers' quality has received limited attention. Therefore, the intersection of the two research streams focused on organizational design and quality management in supply chains provides an opportunity for a new line of inquiry. One mechanism through which buyer-side organizational structure changes can influence suppliers is performance signaling, in which buyers routinely send implicit or explicit performance signals by focusing on priorities such as quality, cost, or delivery, or by conveying a combination of these. When such signals are inconsistent or competing, they can create ambiguity and hinder supplier alignment. These signals are often shaped by the buyer's own organizational structure and incentives, which, in turn, influence supplier decision-making.

While both organizational structure and non-financial incentives can independently influence individual behavior and performance, they also interact in important ways. Prior research has shown that structure can shape how incentive mechanisms are perceived and acted upon. For example, Joseph et al. (2016) find that the organizational structure influences the interpretation of performance feedback, which in turn affects product phase-out decisions and market performance. Similarly, Marengo and Pasquali (2012) argue that structure and incentives jointly affect organizational control and learning. When employees are incentivized to pursue long-term improvement initiatives that involve experimentation, a decentralized and flexible structure with greater autonomy may be necessary to allow effective execution (Lambert et al. 1993; Karpoff 2001; Li et al. 2021; Natter et al. 2001). This perspective aligns with broader work that emphasizes how structure and incentives together shape decision making and learning within organizations (Harris and Raviv 2002; Nadler et al. 1997).

Our study uses empirical data to contribute to organizational theory and operations management, specifically quality management in supply chains, by examining how knowledge-based mechanisms of governance interact in the supply chain arena

and support quality performance at suppliers. A majority of the studies in this stream are analytical in nature and explore aspects of buyer-supplier relationships such as optimal cost sharing in supply chains (e.g., Chao et al. 2009; Cao et al. 2020). Some empirical studies have analyzed quality management in buyer-supplier contexts related to the effect of vertical integration (Pierce 2012), failure penalty (Handley and Gray 2013), learning by doing (Ittner et al. 2001), and forgetting (Agrawal et al. 2020). Our paper contributes to this broad stream of literature by analyzing the role of organizational design, specifically the joint effect of incentives and structure, on the quality of sourced components over time. Greenwood and Miller (2010) called for renewed attention to issues relating to organization design (see also Burton et al. 2006). In a similar vein, Roth et al. (2016) have stressed the need to better understand the context and sources of knowledge creation in supply chains from a practice perspective through improved collaboration and coordination with industry and analyzing real-life contexts of supply chain knowledge creation (p. 1486). Our paper addresses this call directly, in that we engage closely with a large organization over 4 years of collaboration, study a longitudinal quasi-experiment, and highlight the mechanisms of knowledge creation and their linkages with organizational structure. The central research question that we examine is: *How do changes in implicit incentives and organizational structure at a buyer influence quality performance at suppliers?*

2 | Literature Review

We review three interrelated streams of literature: the role of incentives in organizational performance, the role of organizational structure, and the interaction between the two. We also draw on research in supply chain quality management to link internal organizational changes with supplier-facing quality outcomes. Together, these streams provide the foundation for our investigation into how a buyer's organizational design can influence supplier performance.

2.1 | Incentives and Organizational Performance

Incentives influence individual behavior by directing attention and effort toward specific outcomes. Financial incentives such as bonuses and stock options are linked to measurable performance and have been shown to improve task completion and output quality. Non-financial incentives such as performance evaluations, job-related goals, recognition, and career advancement also shape behavior, especially in settings that require sustained engagement and collaboration (De Leeuw and Van Den Berg 2011; da Silveira et al. 2013; Steinbach et al. 2017; Cadsby et al. 2007).

Clear performance goals act as implicit incentives by defining what matters. In supply chain contexts, such targets can motivate quality engineers of suppliers to pursue long-term improvement initiatives, including sustained collaboration with suppliers, root-cause analysis, and the exploration of systemic solutions. These efforts contribute to the creation of quality knowledge, which includes codified elements such as documented standards and specifications, as well as tacit elements

such as problem solving know-how and contextual understanding developed through repeated interaction (Gambardella et al. 2015). Unlike codified knowledge, tacit insights are harder to transfer without direct engagement and often depend on the autonomy and continuity afforded by organizational structure, a point we revisit in later sections.

Although prior research has focused on the role of incentives in driving internal performance, less is known about how buyer incentive systems affect supplier behavior and quality outcomes (Lee and Meyer-Doyle 2017). Our study examines this external effect, focusing on how non-financial, goal-based incentives, when combined with appropriate organizational structure, can influence supplier-side quality performance.

2.2 | Organizational Structure and Organizational Performance

Early research on organizational structure dates back to Max Weber's theory of bureaucratic organization with clearly defined roles, division of activities, and authority allocation (Sager and Rosser 2009; Constanas 1958; Sine et al. 2006). Organizational structure shapes how decisions are made, how tasks are divided, and how authority flows within firms. A fundamental distinction in this literature is between mechanistic and organic structures (Burns and Stalker 1994). Mechanistic structures emphasize centralization, formal rules, and clear hierarchies. Organic structures, in contrast, are more decentralized and flexible, enabling greater responsiveness and open communication.

Building on this foundation, scholars have examined how structure affects learning, control, and adaptation in complex environments (Mintzberg 1989; Burton and Obel 2018; Nadler et al. 1997). Organic structures are particularly suited to tasks that require discretion, collaboration, and iterative problem solving. Mechanistic designs are more effective in stable settings with predictable tasks. Structural choices influence how organizations balance efficiency and responsiveness.

In supply chain contexts, organizational structure affects how buyers interact with suppliers. Decentralized organizational structures allow better identification of problems, exploration of solutions, and coordination of improvements directly with suppliers. This autonomy is important in quality management, where responsiveness and local problem solving are often necessary for sustained improvement (Lambert et al. 1993; Zhang and Sun 2023). Highly formalized structures may limit such engagement through approval requirements or constrained discretion.

Organizational structure also influences how performance expectations are communicated externally. As noted by Pierce (2012), a buyer's organizational form shapes how suppliers interpret its priorities regarding quality, cost, or delivery. These signals are conveyed through reporting lines, role definitions, and coordination routines. Misalignment between structure and stated priorities can lead to mixed signals, reducing supplier clarity and undermining performance alignment.

In sum, organizational structure serves both an internal coordination role and an external signaling function. In what follows,

we explore its interaction with incentive systems to further examine how organizations and their partners respond to strategic goals.

2.3 | Interplay of Organizational Structure and Incentives in Shaping Performance

Scholars have long emphasized that incentives and organizational structure are interdependent elements of organizational design (Marengo and Pasquali 2012; Zhang and Sun 2023; Karpoff 2001; Li et al. 2021; Stiglitz 1989). Burton and Obel (2018) highlighted that organizational performance often includes incentive systems, along with other design parameters such as communication flows, leadership practices, and information systems. The notion of fit between an organization's structure and its incentive system has been extensively examined in literature on organizational design and contingency theory (Tushman and Nadler 1978; Galbraith 1974; Donaldson 2006; Huber and McDaniel 1986; Roberts 2007).

Empirical studies have supported this interdependence. Karpoff (2001) demonstrated that the joint alignment of incentives and structure improves outcomes in explorative project environments. Similarly, Zhang and Sun (2023) found that the ambidexterity of organizational innovation benefits from the interaction between decentralized structures and appropriately calibrated incentive systems. The underlying logic is that structure governs how information flows and how decision rights are allocated, while incentives direct attention, motivate behaviors, and shape how knowledge is generated and applied (Stiglitz 1989; Westphal and Zajac 2013; Marengo and Pasquali 2012).

This interaction becomes especially relevant in environments that require autonomy and discretion. Organizational structures distribute roles, responsibilities, and authority, which implicitly shape non-financial incentives such as task ownership and visibility. As Marengo and Pasquali (2012) noted, decision rights should ideally be placed with those closest to relevant knowledge, particularly when knowledge is tacit or difficult to transmit. However, delegation of authority introduces risks if local agents pursue their own interests rather than those of the organization (Jensen and Meckling 2009). This tradeoff creates the need for a balance between local autonomy and organizational control (Aghion and Tirole 1997), which incentive systems help manage by aligning individual effort with organizational goals.

In contrast, incentive systems designed to support long-term exploratory behavior also require appropriate structural supports. Agent behavior is often shaped not only by actual goals but also by how evaluation mechanisms are perceived. Thus, structure and incentives must be coordinated to support task execution, learning, and innovation.

Although the interdependence of organizational structure and incentives has been studied in internal contexts, their joint effect on supply chain quality remains underexplored. Specifically, how the incentive design of a buyer and structural choices jointly influence supplier behavior and quality outcomes has received little empirical attention. This paper addresses this gap by examining how buyer-side changes

in implicit incentives and organizational structure interact to affect the quality of sourced components from external suppliers.

2.4 | Quality Management in Supply Chains

An important literature stream relevant to our study is that on quality management, rooted in the ideas of Deming (1986), Juran (1951), Crosby (1967), and Ishikawa (1985). These works emphasize that quality improvement is inherently long-term and that better process quality enhances product quality, operational performance, customer satisfaction, and market share (Garvin 1984).

Within this tradition, Mukherjee et al. (1998) explore how Total Quality Management (TQM) facilitates learning and the creation of technological knowledge. The adoption of TQM is often accompanied by changes in organizational design, such as lean systems (Staats et al. 2011). Other studies assess the role of formal certifications (Levine and Toffel 2010) and organizational alignment in the generation of quality knowledge (Gokpinar et al. 2010).

A parallel stream uses analytical models to examine cost sharing (Chao et al. 2009), contracting (Balachandran and Radhakrishnan 2005), order quantity (Dai et al. 2012), quality investment under recall risk (Dong et al. 2020), and information acquisition (Cao et al. 2020). Empirical studies investigate quality management in buyer–supplier contexts, focusing on vertical integration (Pierce 2012), penalties (Handley and Gray 2013), learning-by-doing (Ittner et al. 2001), and forgetting (Agrawal et al. 2020). The latter highlights that quality knowledge sourced externally is more vulnerable to decay than that generated internally.

Other studies explore mechanisms through which buyers improve supplier quality: performance monitoring (Gualandris et al. 2015; Dong et al. 2016), contracts and audits (Chen et al. 2008, 2020; Ha et al. 2023; Bondareva and Pinker 2019; de Zegher et al. 2019), and supplier engagement through long-term partnerships (Modi and Mabert 2007; Yeung 2008; Lee and Li 2018; Klassen and Vachon 2003; Sroufe and Curkovic 2008; Lu and Tomlin 2022). Trent and Monczka (1999) classify quality management practices by procurement stage, complexity, and expected improvement. They suggested that involving suppliers in joint development and problem solving can improve outcomes. Lascelles and Dale (1989) emphasize that communication gaps, weak programs, and lack of commitment hinder quality partnerships.

Dyer (1996) argues that buyer investments in suppliers foster information sharing, but may raise transaction costs. Repeated trust-based interactions (Helper and Sako 1995) can produce productivity gains (Dyer 1997) and reduce long-term costs (Nishiguchi 1994). Such relationships also influence new product development (Clark and Fujimoto 1991), complexity reduction (Bensaou and Anderson 1999), managerial attention (Dhanorkar et al. 2018), and task interdependence (Ramasubbu et al. 2019).

A related stream focuses on the creation and transfer of quality knowledge, defined as both explicit (e.g., standards, procedures) and tacit (e.g., experiential insights from collaboration). Knowledge transfer is shaped by complexity and context (Szulanski 1996). Tacit knowledge is more difficult to share than codified knowledge (Nonaka 1994), and absorptive capacity (Cohen and Levinthal 1990) is essential for knowledge assimilation. Better sourcing knowledge reduces manufacturing complexity (Novak and Eppinger 2001).

Although prior work emphasizes contracts, audits, and collaboration, few studies examine how a buyer's internal design, particularly its organizational structure and implicit incentives, shape supplier quality. Balachandran and Radhakrishnan (2005) are a rare exception, showing that centralized R&D can support knowledge transfer in buyer–supplier settings. Our study builds on this insight by analyzing how a buyer's organizational structure and implicit incentives jointly influence quality outcomes at suppliers. This addresses a significant gap in understanding how organizational structure and implicit incentives support quality governance across firm boundaries.

3 | Research Context—Field Study

Auto Inc. produces cars and SUVs/light trucks in separate plants (Car and Truck). The two units, Car and Truck, are managed as distinct entities with independent management for each. As a result of an internal managerial decision at Car, the firm experimented with redesigning the roles, impacting how organizational agents are evaluated and recognized, and the organizational structure in its car plant, adding a unit specifically focused on improving quality at its suppliers, called the Supplier Improvement Unit (SIU). The managers at Truck decided to maintain their existing organizational structure. Since both plants depended on numerous suppliers, many of which provided components to both plants, the change in organizational structure at Car provides an interesting research opportunity: a longitudinal study of the process of organizational refocusing on quality in Car and its effect on the quality of components sourced by both plants. Figure 1A presents the organizational structure employed from 1997 to 2007 at the two plants.

3.1 | Creation of the Supplier Improvement Unit—Improvement via a More Organic Structure

In 2007, Car created the SIU, comprising 25 engineers drawn from the existing Car unit. The primary job of the new SIU engineers was to improve the quality of the sourced components by working almost full-time to solve problems in the supplier plants. Two important aspects of the SIU deployment are: (a) no counterpart to Car's SIU was created in Truck, and (b) within Car, the SIU did not report to the operations head but instead to the plant manager (Figure 1B). The SIU engineers were focused on differentiated tasks of their assigned vendors and worked to resolve individual quality issues with empowered decision making. Although the SIU was a new unit in Car, other Car

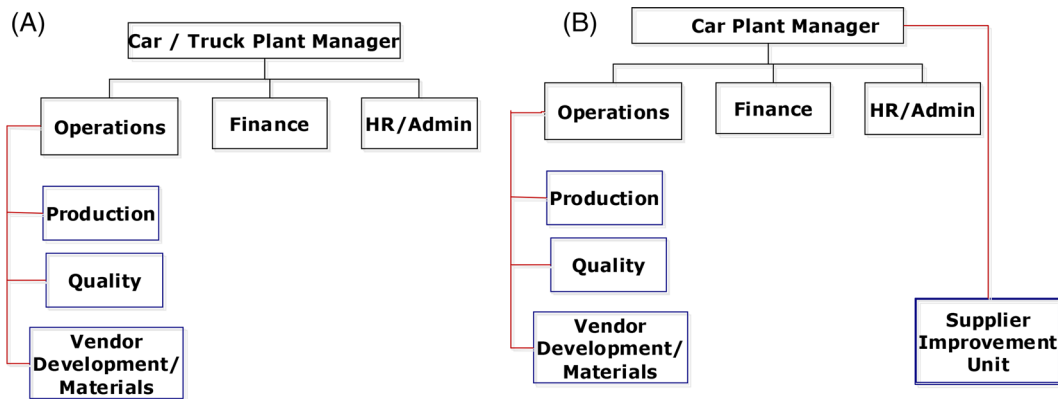


FIGURE 1 | (A) The mechanistic organizational structure at AUTO before 2007 in Car and Truck. (B) The more organic structure deployed in Car after 2007 (the Truck structure remained as illustrated in A). Initially (as in (A)), at both Car and Truck, the quality and vendor development departments reported to the operations head. The vendor development department dealt with developing new components, negotiating prices with suppliers, and day-to-day sourcing, logistics, and supply to production lines, but had shared responsibility for quality. The quality department at each plant was responsible solely for inspecting and accepting parts shipped by the suppliers; it had no role in component design at Auto Inc. or production decisions at Auto's suppliers. Approximately 20% of engineers from vendor development and quality departments at Car were transferred to the Supplier Improvement Unit (SIU).

and Truck processes, including their procurement processes, remained identical. The creation of the SIU and the change in the reporting structure jointly created implicit incentives and structural decision authority for SIU engineers to focus on long-term quality improvement and knowledge creation within the supply chain, rather than on short-term cost and delivery objectives.

Our discussions with Car and Truck engineers and the Car plant manager revealed several features that indicate that the SIU, as deployed, could be classified as an organizational redesign that combined long-term quality incentives and a relatively more organic structure. Therefore, the SIU creation and change in the reporting structure combine changes in incentives, specifically non-financial implicit incentives, and organizational structure. The Car plant manager wanted to improve the quality of sourced components. However, he faced three main problems. First, there was ambiguity around the notion of quality improvement: a very large number of potential quality problems were coming up in daily reviews, but it was not clear how these quality problems were to be resolved. Second, frequent interdepartmental conflicts erupted over which of the quality problems were the most critical and who should be responsible for solving them. Third, the plant-level engineers and operating managers were focused primarily on quantity (volume of production) and on-time deliveries—objectives that the plant manager classified as “short-term” or “tactica.” However, the quality of the components sourced affects current and future profitability. In addition, quality problems may take time to arise, and long-term corrective actions often involve exploration of new quality and technology knowledge. Therefore, quality-related decisions can be considered to be more “strategic” and long-term in nature. The plant manager stated that it was very difficult to focus people on long-term quality objectives, particularly when they were required to perform both the day-to-day operational tasks focused on timely delivery of products to the market and engage in root-cause analysis and explore solutions for long-term resolution of the quality problems at the suppliers' manufacturing facilities.

Long-term quality improvement actions included a systematic identification of quality issues, performing a detailed root-cause analysis of the problems, evaluation of improvement needs, and execution of joint quality improvement projects with supplier manufacturing teams. The primary distinction of the SIU structure was its sole focus on quality improvement at the supplier level, unlike the broader responsibilities of the supplier management team, which included development, delivery, cost, and quality, without exclusive emphasis on quality improvement.

When Car created the SIU, each SIU engineer was given direct responsibility for the quality of components sourced from approximately 10–14 suppliers. This matching of engineers to suppliers was done within SIU itself and was based on the technology of the sourced components; for example, suppliers of plastic components were handled by one SIU engineer, while another looked after castings and forgings. Such an arrangement called for increased face-to-face communication with vendors as well as teamwork with other SIU engineers; a contributive approach to problem solving of this sort is a tenet of organic structure (Burns and Stalker 1994). All SIU engineers reported to a single SIU manager, who reported to the plant manager. The SIU's reporting structure was significantly flatter compared to the operational quality control team's structure. Specifically, the SIU bypassed intermediate layers, such as the vendor development manager and the operations manager, which were part of the mechanistic reporting structure in the quality control teams at both Car and Truck. For example, a typical engineer's reporting in the vendor development department would be as follows: Engineer (reports to) → Group Head {Metallurgy/Electrical/Mechanical} (reports to) → Vendor Development Manager (reports to) → Operations Head (reports to) → Plant Manager. Contrast this with the SIU structure, wherein the reporting was flatter: SIU Engineer (reports to) → SIU Manager (reports to) → Plant Manager.

The SIU engineers had broad-based responsibilities—they could work on process improvement at suppliers, could liaise

with Car's design department for changes in drawings and design, could ask suppliers for investments if needed, and could also work on Car assembly lines to solve problems. Their involvement in assembly line troubleshooting was limited to specific instances where supplier-driven issues surfaced during production, requiring immediate coordination with Car's design or operations teams. These interventions were focused on identifying root causes and implementing lasting solutions, reflecting the SIU's mandate for systemic quality improvement.

The SIU complemented Car's short-term focus on addressing immediate operational challenges and ensuring production efficiency. However, prior to the SIU, Car's short-term focus often defaulted to immediate firefighting rather than structured quality improvements. By addressing chronic, long-term issues through systemic problem-solving, the SIU bridged this gap, indirectly benefiting cost, delivery, and operational metrics. Thus, the SIU balanced immediate operational demands with long-term quality objectives, fostering a more integrated and sustainable approach to quality management.

The SIU's flexibility allowed it to address chronic issues affecting both supplier processes and Car's operations. Its direct reporting to the plant manager ensured a focus on long-term quality improvements, even during occasional on-site assembly interventions. This autonomy and broad scope of responsibilities, characterized by minimal rules and independent decision-making, epitomized the organic structure of the SIU (Douglas and Judge 2001). In the following section, we describe the two key mechanisms through which this new organizational design comprising the change in task assignment and goal, and the reporting structure facilitated quality improvements.

3.2 | Long-Term Incentives Within the Buyer

SIU engineers did not have responsibility for ensuring the short-term goals of production; their primary responsibility was to improve the quality of components from their assigned sources, and each SIU engineer's yearly performance evaluation was dependent on year-to-year quality improvements at those suppliers. In contrast, monthly production targets and daily monitoring of "achievements till date" for these monthly targets were deployed for all other groups under the operations head. Creation of the SIU aided in switching to long-term quality improvement, since it made it possible to deploy a new performance measurement system at the buyer end. The plant manager could provide resources and support to the SIU engineers so that they were aligned and committed to long-term quality improvement, which was not possible earlier. An SIU engineer said about the difference between the two scenarios:

When I was working in the quality department, we were focused on day-to-day quality. Our time was spent in segregating bad components from good ones, to ensure production. Many times, because of components that did not fit, or because of components that were not of the required quality, the production

was stopped. Then, the quality department was blamed for the loss of production. I remember many unpleasant exchanges with the vendors, as well as my colleagues in other departments, such as materials and production. There was also high inventory, and defective components kept on reappearing despite follow-ups with the vendors. Of course, I and other engineers became very proficient in responding to emergencies: how to quickly separate bad components, how to deflect inquiries, and how to put blame on the vendors or on the designers. Today, my time is spent working on solutions that may take time to implement (such as engineering changes, poka-yoke, or facility layout changes), and which are almost always unique for a supplier. I do not spend my time on day-to-day operational problems. Things have improved—not only in incoming quality and inventory but also in social interactions. There is less of a load on my colleagues in production and quality—it is a superior way of working. Of course, it is not all rosy. There are still tough moments, but I think the change in structure has made a big difference for all of us.

The engineer's quote enables us to understand that the creation of the SIU in the Car plant changed the way engineers were working, by focusing them on long-term solutions to chronic problems. The SIU team gave the plant manager a way to focus on longer-term results, which had not happened when quality reported to the operations head. Engineers who transferred to the SIU from other groups received no changes in salary but were freed from the day-to-day firefighting mode, solving today's problems. They could concentrate on their new goal (of improving quality at their assigned suppliers), even if the improvement of quality took time. The plant manager observed:

Yes, it was a leap of faith, possibly. However, it was clear that we had to change the focus on quality and handle it separately if we wanted results. The SIU reporting change helped this change in focus. The change was subtle but effective.

Yearly targets and yearly reviews helped create a long-term focus on improvements. Therefore, the criterion and frequency with which the SIU engineers were evaluated and performance feedback provided changed significantly from operational quality assurance to long-term quality improvement at the supplier facilities. This change in non-financial incentive mechanism allowed the SIU engineers to devote their effort and attention to long-term quality issues and resolution. The SIU engineers could solve problems at the right place, even if it took more time. Each engineer could focus on differentiated expertise, unique to their assigned suppliers. These policy changes helped create long-term incentives for the SIU engineers, who then solved the most important problems at their source, usually the supplier end. Such solutions reduced

recurrences. Slowly, the quality of the sourced components improved. Physically, the engineers transferred to SIU remained on the same floor at Car, though separated in a dedicated space: indeed, some SIU engineers did not need to move at all. However, their day-to-day work changed. They then spent their time primarily at the supplier plants, often visiting multiple suppliers every day. Before the SIU was formed, many of these engineers were employed in tasks such as separating good components from bad ones, and trying frantic manipulations on the assembly line, so that (somehow!) today's cars could be made. At first, these firefighting tasks were partly assumed by colleagues, with SIU engineers continuing to perform them for supplies from the specific vendors that they were responsible for. That changed slowly as the SIU engineers zeroed in on and arrested more and more problems at the vendor end rather than at Car's assembly line. Back at the Car, jobs and tasks related to quality were gradually reduced or totally eliminated. Much of the work eliminated had previously been performed by the very engineers now working within the SIU. Not only were fewer defective parts produced, and fewer resources devoted to non-value added work (e.g., inspection, rejection, and disposal of defective parts), but SIU engineers found their own work more rewarding and their relationships with personnel at the suppliers much improved (See also Oliva and Sterman 2001).

Note that the contractual incentives for suppliers to Car and Truck remained unchanged by the implementation of Car's SIU. Those incentives are related to three aspects of sourcing. First, the supplier not only is not paid for components that are rejected, but also bears the costs of removing the rejected components and disposing of them. High defect rates are thus costly to any supplier. Second, Auto Inc. awards a higher share of ongoing business to suppliers whose quality has been higher (in both Car and Truck). Third, when developing new models (in both Car and Truck), Auto Inc. routinely chooses suppliers who have demonstrated higher quality (as measured by ppm defects) in supplying components for previous models. Therefore, in the short term as well as the long term, before and after the implementation of the SIU, all suppliers have had incentives to improve quality.

The natural question is: Why were quality problems not being addressed satisfactorily prior to the creation of the SIU? We suggest that (i) most of the persisting quality problems were complex, and (ii) there was considerable ambiguity around issues of quality measurement and improvement. The knowledge of Car's processes and design required to make these changes was the knowledge that suppliers did not have prior to their collaboration with SIU engineers.

Consider, for example, a problem encountered by Car in its assembly line that appeared to be caused by a particular component supplied by a particular supplier. Was the problem actually with the supplied component, or caused by components supplied by other suppliers (which interacted with the focal component in question), or in the design of the component, or in Car's assembly process? Causal ambiguity around the actual nature of a problem obviously impeded improvement, not to mention a definitive solution to the problem (Lippman and Rumelt 1982). Prior to active buyer-supplier involvement, it had not been

possible to reduce the ambiguity around the root causes of such problems; the problems were not researched deeply, root causes were not pinpointed, and thus all solutions were ad hoc. A quote from a supplier of components to both Car and Truck is helpful in understanding that problem solving under the earlier organizational structure focused only on quick fixes:

Earlier in Car, just as is true many times even today in Truck, if there was a problem in production, and my component was involved, I would be called up and shouted at. Then, I would attempt to do a quick fix. What was important was to solve the immediate issue, by any means, to ensure that production for that day was not affected. Today, things have changed. We have the SIU engineer in Car, whose sole responsibility is to lower my defect level. Now, every time my component fails, my defect level increases—but then the SIU engineer gets into the action, goes to the root cause, and we solve the problem at the right place. It takes time. But the point is—we get new problems in Car, but not old ones.

The SIU's structure enabled it to address both simple and complex quality issues effectively. For simpler issues, the SIU facilitated immediate problem-solving through targeted interventions, ensuring operational continuity. For more complex and ambiguous problems, such as those involving multiple interacting components or the ambiguous nature of defects, the SIU engaged in a detailed root cause analysis and Pareto evaluations. This dual capability highlights the SIU's balanced approach to resolving immediate production challenges while fostering long-term quality improvements, a critical element differentiating it from the pre-SIU organizational structure.

Our study of typical SIU improvements was instructive (Online Appendix B provides several examples). The engineers went into classic problem-solving detail by asking five Whys, doing a Pareto analysis of defects, and developing Fishbone diagrams. The supplier-side operators and engineers were also involved in this work. Thus, the following sequence of actions was visible: (i) the buyer and supplier observe and assess what is happening with respect to quality, (ii) they design a conceptual response to quality problems, (iii) they test the response, and (iv) they make subsequent corrections accordingly. This sequence of actions leads to new knowledge about root causes (know-why) in addition to operational knowledge (know-how). Know-why differs from know-how in that the focus is on understanding why the problems are occurring in a process, by researching more deeply, as against the simple understanding of how the day-to-day work is being done (Lapr   et al. 2000).

3.3 | Aligning Organization Structure and Incentives to Enable Decentralized Quality Decisions

The creation of the SIU introduced two significant shifts. First, a reorientation of non-financial implicit incentives toward long-term quality improvement and second, a transition to a more

organic structure, enabling SIU engineers to act with greater independence and discretion. Before the SIU, engineers faced multiple layers of hierarchy to secure approvals for supplier-facing initiatives. The new SIU structure streamlined these processes by allowing direct reporting to the plant manager.

This organizational structure change provided three key advantages. First, supplier improvement efforts often required new investments in equipment or processes that required approvals or financial discretion from the buyer side that were not included in standard sourcing contracts. With easier access to strategic leadership, SIU engineers could now obtain faster approvals and initiate necessary changes. Second, reporting to the plant manager ensured that long-term initiatives were not deprioritized by operational constraints. Third, the SIU structure improved cross-functional collaboration because the new reporting structure signaled the strategic importance of the SIU's work to others in the organization. SIU engineers could directly approach departments such as design, finance, and tooling without routing requests through the operations chain. Departmental heads were more responsive to requests from SIU engineers, recognizing their alignment with the plant manager's priorities. This shift supported more coordinated and sustained problem-solving at the supplier end.

The access of SIU engineers to top leadership, coupled with their clear mandate, fostered faster and deeper engagement in supplier improvement projects. A quote from one of the SIU engineers illustrates the above insights better.

Earlier, getting approvals for investments in jigs, fixtures, new machines, or processes would need us to go through the ranks of the organizational hierarchy to get any approval. I would need to submit a proposal to my manager, who would pass it on to the operations manager, who would then pass it on to the plant manager for approval and financial provisioning. However, now we can directly reach out to the plant manager for easier and quicker approval. Similarly, earlier, engaging design or finance would need us to go through the same organizational hierarchy, where we would need to depend on the operations manager to request support from other departments. However, now we can directly reach out to other departments and easily coordinate. Knowing that we directly report to the plant manager, other departmental heads now easily respond to our support requests. There is a definite change in how I and others work for approvals now, and indeed it helps us to focus more on the real problems at the supplier.

Thus, the creation of the SIU involved two interconnected changes: first, a shift in the implicit non-financial incentives for SIU engineers due to changes in their roles and performance targets; and second, a transition to a more organic organizational structure. It is important to note that the rest of the engineers and departments retained their previous mechanistic structure while the SIU operated with greater flexibility, autonomy, and

cross-functional reach. Car thus evolved into a hybrid system, combining organic elements within the SIU alongside the existing mechanistic model. This hybrid form enabled the buyer to better support long-term supplier-facing improvements.

4 | Hypotheses

In this section, we use the existing literature to develop hypotheses related to quality knowledge and the spillover of quality knowledge generation to other buyers as a result of buyer-side changes in organizational structure. We develop the hypotheses from a theoretical perspective for generalizability and then align them with the field study of Auto Inc., which helps contextualize the hypotheses at an operational level and provides practical validation of the theoretical understandings.

4.1 | Quality Improvement Effects of Change in Incentives and Organizational Structure

Galbraith (1974) defined organizational design as the alignment of objectives, task distribution, individual actions, and collaboration across units. This design is influenced by how firms process information internally (Daft and Lengel 1986; Sosa et al. 2013) and shapes the behavior and activities of organizational agents (Gokpinar et al. 2010). It encompasses who makes decisions, how communication is structured, how individuals allocate attention and effort, and how the organization relates to its external environment (Csaszar 2013; Child 1973; March and Simon 1958; Cyert and March 1959; Rowan 1982; Leifer and Huber 1977). Across these dimensions, incentives and structure emerge as two interdependent levers that jointly guide behavior and shape organizational performance (Burton and Obel 2018; Keidel 1994; Tushman and Nadler 1978; Huber and McDaniel 1986).

Organizational incentives help in goal and task alignment (Kretschmer and Puranam 2008), exploration of new knowledge or exploitation of existing knowledge (Lee and Meyer-Doyle 2017), performance motivation (Gambardella et al. 2015), and long-term risk-taking behavior (Wright et al. 2007). In the context of the SIU creation, changes in indirect non-financial incentives implicitly emerged from the changes in role and performance targets. Engineers tasked with both daily quality assurance and long-term improvement often end up prioritizing immediate operational issues due to several implicit factors. Urgent problems, such as defects or customer complaints, demand quick responses and are often met with recognition or avoidance of escalation, subtly reinforcing this behavior. These tasks are also easier to measure, such as the number of issues resolved or response time, providing instant feedback and a clear sense of accomplishment. In contrast, long-term quality initiatives involve uncertainty, take longer to show results, and are harder to evaluate, making them less attractive. Moreover, these strategic projects often involve experimentation and risk, which engineers may avoid to maintain stable performance. With limited time and attention, the conflict between immediate and strategic responsibilities leads engineers to focus on what is urgent and clearly rewarded, sidelining longer-term improvement efforts.

By creating a separate group of SIU engineers solely focused on long-term quality improvement of supplied parts and components, the firm redesigned the role to serve as an implicit non-financial incentive in a way that inherently encouraged exploration, strategic thinking, and innovation. This clarity of the role eliminated the confusion of balancing immediate operational tasks with forward-looking initiatives, allowing engineers to fully commit to long-term goals. Freed from the constant demands of urgent day-to-day operational issues, they gained the time and mental space needed for root cause analysis, experimentation, and creative problem solving through exploration of new knowledge and technologies. The dedicated nature of the role also reduced the fear of failure, signaling that trial and error is not only accepted but expected. In addition, being part of a specialized unit enhanced their sense of autonomy, status, and connection to senior leadership, factors that naturally boost motivation and ownership. Placing this team within a more flexible, strategy-oriented structure further ensured that their efforts aligned with the organization's broader goals, rather than being pulled into short-term operational pressures. The changed role was supported by the change in the organization structure to a more organic one for the SIU engineers.

Prior literature has established that organizational structures vary in their support for exploration versus exploitation. Organic structures, with their flexible roles and minimal formalization, are more conducive to exploration and innovation, while mechanistic structures support standardization and control (Burns and Stalker 1994; Csaszar 2013; Lavie et al. 2010). We build on this foundation to examine how the adoption of a more organic structure, when combined with role-based incentives, supports long-term quality improvement in supply chains.

To quote Lavie et al. (2010), “These alternative structures can correspondingly facilitate exploitation or exploration. Exploration entails non-routine problem solving and search for new knowledge that may make information processing inefficient under centralized decision making. In turn, formalization is expected to constrain exploration and facilitate exploitation through incremental improvements in processes and products” (p.122–123). This distinction has been reinforced by subsequent research, which shows that structural flexibility promotes exploration, while formalization and hierarchy support control and standardization (Cardinale 2018; Dougherty 2001).

Often such actions can materialize and manifest in long-term improvements, but are fraught with time and outcome uncertainty (Wiseman and Gomez-Mejia 1998). In the context of supply chains, buyer firms can create incentives for their agents to focus on quality improvement and the generation of quality knowledge at supplier firms. In an organic structure, frequent and open communication that cuts across organizational hierarchies and the ability to provide synchronous feedback (Daft and Lengel 1986) can enhance the ability of the supplier to learn from the buyer (Lane and Lubatkin 1998). Thus, the choice of actions by buyers' agents can provide signals to supplier organizations to undertake long-term improvement actions rather than simply ensuring that short-term targets (such as cost and delivery) are met. Organizational goals associated with outcomes that are clearly measurable and have a relatively direct causal association with efforts tend to motivate managers more than

when the outcomes are uncertain or the causation is confounding (Wood and Bandura 1989). Dougherty (2001) suggested that differentiation of activities supports focused goal formation, which, in turn, helps organizational agents value the strategic and transformative nature of work; such activities are “not just more of some aspects of mechanistic organizing and less of others (e.g., more coordinating devices, fewer levels), but comprise a very different way of imagining work” (p. 628). In practice, organizations often require a balance between exploration and exploitation. Similarly, Marengo and Pasquali (2012) suggest that organizational design allocates authority and implicit incentives that complement each other in “non-trivial” ways and must be tuned to achieve long-term objectives. In general, adjusting implicit incentives through role and task design, especially within more organic organizational structures, can foster independent decision-making and support long-term quality improvement, especially in uncertain and innovation-intensive environments.

We posit the following hypothesis that is grounded in our field context, where the creation of the SIU at Car through dedicated roles, changed incentives, and a shift to a more organic structure enabled the sustained improvement of component quality, exemplifying how aligned design choices can drive outcomes.

Hypothesis 1. *The creation of a dedicated team focused solely on supply chain quality improvement, coupled with a shift to a more organic structure within a mechanistic buyer organization, is positively associated with improved supplier component quality.*

4.2 | Factors Influencing Quality Improvement—Technological Complexity, Number of Parts Supplied, and Number of Manufacturing Locations

While the creation of a dedicated team and the adoption of a more organic structure can improve supplier quality, the extent of these improvements may depend on specific part and supplier level factors. Drawing on field insights and relevant literature, we identify three such moderating factors: (i) the technological complexity of the supplied part, (ii) the number of different parts a supplier provides, and (iii) the number of manufacturing locations involved in producing a part. Each of these factors affects how (i.e., in what way) quality knowledge can be generated, transferred, and implemented across a supplier's operations.

As discussed earlier, the SIU team's new role and structure enabled focused exploration of long-term quality issues at supplier sites. However, the effectiveness of such efforts may vary based on the nature of the product and process involved. Components that are technologically complex or require coordination across many parts or plants often involve deeper search, higher uncertainty, and longer resolution cycles. These conditions may influence how fully quality knowledge is created and translated into improvement.

Product complexity is an important dimension that affects the generation of product knowledge and the transfer of knowledge between organizations (Sousa and Voss 2001; Fynes and Voss 2001). Complex knowledge (i.e., knowledge that involves a comparatively large number of critical and interacting elements) is more difficult to transfer between buyers and suppliers

(Zander and Kogut 1995) than simpler knowledge. For a simple product, successful supply can be achieved with relatively simple rules and pre-planning and, therefore, requires limited supplier-specialized investments (Bensaou and Anderson 1999). But as technology becomes more complex, buyers and suppliers need tighter coordination mechanisms and must engage in more extensive search processes to arrive at solutions to quality problems (Novak and Eppinger 2001). Although one would expect that strong relational ties in the new structure would enhance the transfer of complex knowledge, we contend that causal ambiguity hinders transfer. Barney (1985) states that in complex, highly inter-dependent technological systems, the establishment of cause-and-effect relationships can be very difficult, and thus the root causes of success and failure can take time and be difficult to determine (Fisher et al. 1999).

In our discussions with SIU and supplier engineers, it emerged that it is often easier to solve quality issues related to technologically simpler components such as internal panels, fenders, and external and internal fittings compared to parts of the gearboxes and engines due to the precision requirements, tighter engineering tolerances, and complexity of interaction that such parts have with other components of the drive line of a car. At the same time, both the SIU engineers and the supplier engineers suggested that with the change in the organizational structure in Car, there was an increased and continuous focus toward quality problems and joint quality issue resolution.

An example of a product with relatively lower complexity that was discussed in detail with SIU engineers was that of door seals, beads, and channels. The specific quality problem was related to inadequate door closure pressure. Door seals and beads were made from rubber extrusion on aluminum or steel channels. The rubber profile consists of a combination of a bulb, which is a circular rubber profile that encapsulates the aluminum or steel channel, and a lip, which is a flat rubber profile that sits on the bulb. The bulb provides the strength of the profile so that the profile does not collapse, and the lip provides the dampening effect to absorb the shutting shock. The shut pressure is determined by the consistency of the thickness of the rubber bulb and the stability of the lip. The problem was the inconsistency of the wall thickness of these two parts, which was related to the wear of the tools and dies used to extrude the rubber profiles. The inconsistent thickness caused misalignment with the center of the extrusion machine. The SIU engineers devised an alignment check fixture that fixed the problem. This project did not require a lot of exploration, effort, or time.

A different project related to the suspension system presented a contrast. Suspension is a complex system consisting of many components and subject to intense forces and dynamics. The problem was related to uneven suspension pressure in one of the axles. This problem can occur due to problems with many parts and components, such as suspension springs, shock absorbers, linkage rods, balancing struts, bushings, joints, and bearings. Understanding the source of the problem and then devising mechanisms that can address suspension quality issues required greater time, effort, and exploration of new methods of manufacturing and assembly, such as assembly fixtures, in-process measurement equipment, in-process alignment fixtures, and checking tools. This involved exploring with several people

and teams, such as car designers, technology providers, and production engineers. Many changes needed to be made in this project, including changes in engineering drawings, process parameters, and process steps. In such a case, the organic structure with a relatively flat hierarchy and independent and flexible decision authority was beneficial, as SIU engineers could explore and decide the courses of action and the path of improvement.

In both the examples above, the creation of the SIU helped in solving the quality issues on a long-term and consistent basis. On the one hand, the shift to a more organic structure encourages exploration and should, in principle, support quality improvements in technologically complex products. On the other hand, such complexity increases the time, effort, and uncertainty required to realize those improvements. Therefore, a priori, it is not obvious in which direction technological complexity will moderate the relationship between SIU creation and quality improvement of sourced products. Our discussions with the SIU engineers and suppliers reveal that often technological complexity manifests the interaction, that is, the quality issues of relatively more technologically complex products are more sticky, and the realization of the benefits is relatively more uncertain. One SIU engineer commented the following related to the moderating effect of technological complexity, “while we put in a significantly greater amount of effort and time for more technologically complex products, we do not immediately realize the benefits as much as that for less technologically complex products.”

Product complexity plays a key role in shaping how quality knowledge is generated and applied. We expect that the more technologically complex a supplied component is, the more difficult it will be to improve its quality. Therefore, we posit the following.

Hypothesis 2a. *Technological complexity of the supplied component is associated with a reduction in the extent to which the creation of a dedicated team focused solely on supply-chain quality improvement, combined with a shift to a more organic structure by a buyer, is associated with improvement in the quality of components at its suppliers.*

Another important factor is the breadth of supply: the number of different parts supplied by a particular supplier. If a supplier is using similar processes and technology for many components supplied to a buyer, then we can expect that knowledge acquired from improvement in one component is likely to be transferred quickly to the production of other components (Darr et al. 1995). A dedicated quality-focused team operating within a more organic structure can facilitate this transfer by enabling deeper exploration and application of new product and process knowledge. In our fieldwork, SIU engineers noted that suppliers often experienced recurring quality issues across multiple components, and solving one underlying issue frequently improved several parts at once.

For example, a pressed component manufacturer explained that several of the body interior panels suffered from inaccuracies in the positions of the notches and pierced holes, leading to issues with final assembly at Car. This was because the fitting of the cover of the ceiling and pillar would require a certain accuracy

of the notches and pierced holes. Before the SIU creation, the supplier would often slightly elongate some of the notches and holes to allow assembly fitment. After the SIU creation, SIU engineers involved tooling engineers from Car, changed the tooling design, and developed flexible and auto-adjusting piercing and trimming interior body components. This solution resulted in solving problems in several parts and components, such as the interior roof liners, the A-pillar, the B-pillar, the C-pillar, the inner fender liners, and the dashboard frame. Another example is that of injection molded components such as instrument trim panels, instrument panels, and frames, where warping and sink marks on surfaces were the issue. The SIU engineers explored the sources of this problem and pinned it down to uneven cooling due to an improper mold design. They then enlisted a specialist tooling design company to redesign the gates and runners of the molds for the dashboard to improve the uniformity of material flow and cooling inside the mold. This improvement also helped to improve the design of the molds and tooling for several other injection molded parts and components. Therefore, we posit:

Hypothesis 2b. *The greater the number of different components a supplier provides to a buyer, the greater the extent to which the buyer's creation of a dedicated team focused solely on supply-chain quality improvement, combined with a shift to a more organic structure, is associated with the improvement in the quality of components at that supplier.*

A third factor that may influence the effectiveness of buyer-led quality improvement is the spatial dispersion of manufacturing activities across a supplier's facilities. Prior research highlights the value of physical proximity: colocated project teams improve coordination and project outcomes (Bardhan et al. 2013), while spatial diversification in manufacturing has been linked to both performance benefits (Lampel and Giachetti 2013) and quality risks, particularly in outsourced settings (Gray et al. 2011). However, the role of manufacturing dispersion in supply chain quality improvement remains underexplored. Our fieldwork suggests that when a part is produced across multiple plants, the generation and implementation of quality knowledge face additional challenges. These challenges arise from two main sources. First, process steps distributed across separate facilities require greater coordination for diagnosis and resolution. An example given by the SIU engineers is that of the gearbox casing manufacturing, wherein casting was performed at one supplier plant while final machining occurred at another. A recurring defect in dimensional accuracy was traced to issues in casting, but resolving it required joint efforts of engineers and operators at both supplier sites. Coordinating across locations led to delays and made the resolution effort more resource intensive. Second, even when similar parts are manufactured in different facilities, solutions developed in one plant do not always transfer easily to others. A case in point was the fabrication of external skin panels by the same supplier in multiple plants. Improvements made to the checking fixtures in one location were not adopted uniformly across the other plants, limiting the diffusion of best practices. These difficulties suggest that spatial dispersion can hinder both the creation and the scaling of quality knowledge.

While, on the one hand, the specific focus on explorative long-term solutions of quality issues brought in by the creation of the

relatively more organic structure in the form of SIU should benefit supplied parts manufactured in multiple locations, on the other hand, spatial dispersion of manufacturing process makes the quality knowledge sticky on specific locations, and the relative benefit may be less realizable when the number of manufacturing locations is relatively higher. One of the SIU engineers responded to a specific query on this aspect as follows "Before the SIU creation, we were too busy with making sure that the assembly line in our plant is fed continuously in an uninterrupted manner, and therefore, we focused primarily on incoming quality checks and the central supplier location. However, now we are able to travel to all supplier locations and address specific issues in all these locations. Yet, we prefer single locations since coordination of quality improvement is so much easier and more effective. Multiple locations make it hard to replicate the quality improvements achieved in one location, although now we are able to do much better on this aspect than earlier." These insights suggest that although a more organic structure supports exploratory quality efforts, its effectiveness may be reduced when manufacturing is fragmented across multiple sites. Accordingly, we propose:

Hypothesis 2c. *The greater the number of manufacturing locations at which a supplier produces components, the less the extent to which a buyer's creation of a dedicated team focused solely on supply-chain quality improvement, combined with a shift to a more organic structure, is associated with improvement in the quality of the components at that supplier.*

Hypotheses 2a and 2b reflect two contrasting forces that can influence the results of buyer-led quality improvement. Technological complexity can hinder the sharing and application of knowledge due to the interdependencies among components and processes. In contrast, when a supplier produces multiple related parts, improvements in one component can more easily spill over to others, facilitating broader knowledge creation and transfer. Hypothesis 2c introduces a third dimension of spatial dispersion which has received limited attention in the supply chain quality literature. It highlights how geographically fragmented production can constrain both the generation and dissemination of quality knowledge. Together, these three factors are not direct drivers of improvement, but they may moderate the effectiveness of organizational design interventions aimed at enhancing supplier quality. In the following sections, we test these interactions and assess their moderating effects empirically.

4.3 | Intrasupplier Spillover

An important aspect of enabling long-term knowledge creation by a buyer at its suppliers is the possibility that this knowledge may spill over to other buyers, including competitors (Agrawal et al. 2016). This raises concerns about how such spillovers occur and under what conditions. Prior literature offers two views on the value appropriated from such spillovers. One view highlights the potential benefits for firms that are not the original investors, including those that are located near sources of knowledge activity or that have overlapping operations (d'Aspremont and Jacquemin 1988; Jaffe et al. 1993; Alcácer and Chung 2007). Related work shows that knowledge can also move within

firms or between affiliated units, often supported by routines, personnel mobility, and informal exchanges (Darr et al. 1995; Eppler et al. 1991; Argote et al. 2000).

The second view emphasizes the challenges of knowledge transfer and the variability in how much knowledge moves across units or firms. Scholars have identified factors such as product similarity (Darr and Kurtzberg 2000), geographic proximity (Gittelman 2007), the recipient's absorptive capacity (Cohen and Levinthal 1990), and the complexity of the underlying knowledge or credibility of the source (Szulanski 1996) as important moderators. Within supply chain literature, Kang et al. (2009) show that suppliers may invest in specific buyers when these investments generate future value through improved relationships with the same or new buyers. However, spillovers are not guaranteed. Suppliers often exhibit differences in quality performance across their customers, suggesting that buyer-specific knowledge advantages may not easily transfer (Muthulingam and Agrawal 2016).

These findings suggest that quality knowledge developed through buyer-initiated improvements can, in some settings, benefit other buyers, even if the originating buyer bears the cost of enabling it. This possibility aligns with broader work on indirect organizational learning, which shows that firms often absorb lessons from the experience of others (Thornton and Thompson 2001). Accordingly, we posit the following:

Hypothesis 3. *The creation of a dedicated team with incentives focused solely on supply chain quality improvement, combined with a shift to a more organic structure at a buyer organization, will be associated with improvement in the quality of similar components that the supplier provides to other buyers.*

This hypothesis builds on Hypothesis 1 by examining whether the structural and incentive changes not only improve quality for the focal buyer but also lead to spillovers that benefit other buyers. The core idea of this hypothesis is similar to that of Muthulingam and Agrawal (2016), but differs in both emphasis and mechanism. While Muthulingam and Agrawal (2016) examine the spillover effects of specific quality improvement projects, our study focuses on how organizational redesign through targeted incentives and structural change can enable more systematic and lasting quality knowledge creation. The SIU structure allowed not only the execution of impactful projects, but also the conditions needed for knowledge to be retained and diffused. Unlike the earlier study, which analyzes project types, we examine how buyer-side organizational design influences broader knowledge flows and supports long-term improvement. We also examine how product and process characteristics, such as product commonality, process commonality, and the number of manufacturing locations, influence the extent of spillover.

4.4 | Factors Affecting Spillovers

In the previous section, we suggested that quality knowledge created at a supplier due to the efforts of one buyer can spill over to other buyers being supplied by the same supplier. However, this spillover may be subject to some influencing factors. We consider three characteristics that could influence spillover: (i)

the number of manufacturing locations of suppliers, (ii) product commonality, and (iii) process commonality. Similar to the development of hypotheses related to the moderating effects of quality knowledge generation earlier, we derive the relevant factors that are associated with the spillover of quality knowledge based on the existing literature and the practical insights gleaned from our field study.

Literature has identified the advantages of colocation in knowledge-sharing environments (Bardhan et al. 2013; Mishra et al. 2020). The proximity of manufacturing locations provides structural opportunities for knowledge sharing and knowledge spillovers. Colocation enables easy communication, coordination, and replication of knowledge. For example, much of the quality knowledge gained through the interaction between SIU engineers and supplier personnel manifests itself in the improvement and development of the processes at the supplier end. Often, this knowledge has a significant tacit component that makes spillover difficult unless processes and personnel are co-located for different products. Our field study reveals that most quality improvements are embodied in process improvements such as improved tooling, process changes, and the development of learning and skills of operators and engineers. The spillover of this knowledge requires close proximity of processes and personnel. Colocated processes may also have shared components such as a shared work force and shared process steps, both of which enable spillover of quality knowledge. Therefore, we posit:

Hypothesis 4a. *The greater the number of manufacturing locations at which a supplier produces components, the lower the extent to which a buyer's creation of a dedicated team focused solely on supply-chain quality improvement, combined with a shift to a more organic structure, is associated with improvement in the quality of components provided to other buyers.*

Argote and Miron-Spektor (2011) suggest that organizational context affects knowledge transfer within and outside organizations. Context refers to the shared components of an organization that facilitate knowledge transfer. Some of the shared components that enable knowledge transfer and spillover of acquired knowledge are colocation (discussed above) and the commonality of products and processes. This view is well supported in the literature (e.g., Posen and Chen 2013; Jacobs and Swink 2011). The commonality of product and process allows firms to integrate the gained knowledge with the existing knowledge infrastructure of a firm, and therefore firms can easily transfer the knowledge gained in one product and process to other similar products and processes. A few examples of product and process commonality between Car and Truck sourcing observed during the field study are pressed components with commonality of the fabrication steps and processes, and head and tail lights with shared components such as wiring harness and lamps. On the other hand, suspension components have very little product or process commonality, with significant differences in structure, configuration, and technologies used in different suspensions. In the latter case, quality knowledge gained by a supplier due to interaction with a buyer may have little relevance for the components supplied to other buyers in the absence of commonalities. One illustrative example of product commonality was the redesigned

door seal with steel mesh support, developed initially for Car to address nonuniform shut pressure and long-term flexibility. Since Truck used similar door seals with comparable design and material characteristics, the improvements made in Car were easily transferred. Similarly, process commonality allowed for the transfer of technical changes. For example, an improved heat treatment protocol developed for gears in Car was readily applied to gears for Truck due to similar metallurgical processes and equipment. When products or processes are shared across platforms or buyers, the application of quality improvements is more straightforward. Accordingly, we posit:

Hypothesis 4b. *The greater the commonality of sourced products between two buyers, the greater the extent to which a buyer's creation of a dedicated team focused solely on supply-chain quality improvement, combined with a shift to a more organic structure, is associated with improvement in the quality of components provided to other buyers.*

Hypothesis 4c. *The greater the commonality of manufacturing processes between two buyers, the greater the extent to which a buyer's creation of a dedicated team focused solely on supply-chain quality improvement, combined with a shift to a more organic structure, is associated with improvement in the quality of components provided to other buyers.*

As with Hypotheses 2a–2c, we test Hypotheses 4a–4c as interactions with knowledge spillover in a moderation setting. We integrate the hypotheses into a research framework shown in Appendix D.

4.5 | Alignment of Research Context With Hypotheses

Before the implementation of Car's SIU, the firm's suppliers did not have much face-to-face communication with Car's buyers, the two parties invested relatively little in the relationship, and the overriding focus at the buyer end was on day-to-day production and fire-fighting. At the buyer end, there was no focus on solving quality problems at the root, at the suppliers. At the supplier end, there was little that could facilitate learning how to improve quality. A quote from a supplier is useful:

Earlier, it did not matter whether there was a problem with my component, if it was Car's fault, or if the problem was with another vendor. We would just segregate the “bad” components and ensure production. Today, if the SIU Engineer tells me about a problem, I know that it is my fault. He goes into details with my workers.

The supplier's quote exemplifies the issue of ambiguity in quality improvement. The focus earlier was on short-term fixes (why a “bad” component is so-labeled is not quite clear). The removal and disposal of components that failed quality standards did nothing to identify why they were defective or how defects might be reduced. Therefore, the problems tended to recur, and the supplier then perceived as more ambiguous what had previously seemed

to be positive outcomes from quick improvement actions. By deploying a new performance management system that involved (a) engineers having direct responsibility and authority to make process and design related changes, and (b) tracking the quality performance over a longer term, Car facilitated long-term learning at its suppliers. The ambiguity around quality issues reduced as buyers and suppliers exchanged more information focused on cause-and-effect relationships, and this increased confidence in the production process (Milliken 1990, also see Thompson 1967, 134): a new learning environment was created. In this environment, the SIU engineer focused on specific problems associated with suppliers allocated to him/her, and the supplier developed root-cause knowledge because of this association with the SIU engineer. As a result, slowly, the focus shifted to longer-term solutions that led to consistent delivery of higher-quality components to Car. Overall, the more organic relationship created by Car triggered a virtuous cycle of quality improvement in its sourcing. The observations presented here suggest that the context of SIU creation is an appropriate empirical setting for our research. As an illustration, we test Hypothesis 1 by demonstrating that the quality of components received by Car, who created the dedicated team with role focused solely on supply-chain quality improvements and whose organizational structure shifted from mechanistic to more organic, improved significantly after the creation of SIU, relative to the quality of components received by Truck, which did not implement such changes.

Table F1 in Appendix E summarizes the exact changes deployed in Car via the creation of the SIU, the now-contrasting approach to sourcing that persisted in Truck, and how differences and their consequences link to extant theory. In this table, we identify the structural elements of an organization in the first column. In the second column, we indicate how the prior structure maps to these elements; in the fifth column, we indicate how a more organic structure maps to these elements. In the third and fourth columns, we indicate the specific policies employed in Truck (and previously in Car), and the specific policy changes deployed in SIU, respectively. Thus, Table F1 links the theoretical constructs of mechanistic and organic structures (Burns and Stalker 1994; Pugh et al. 1969; Adler and Borys 1996; Douglas and Judge 2001) to the SIU intervention.

5 | Data and Methodology

5.1 | Data Collection

Our involvement with Auto Inc. included interactions with the top management team of Auto Inc., interviews with SIU engineers who worked with vendors on quality improvement projects, interviews with quality and vendor development engineers of Truck, involvement in several quality problem identification and problem resolution initiatives, visits to vendors, and interviews of vendor management teams. We were able to work directly with 19 engineers and 102 vendors of Car over the course of 4 years to understand the effects of the organizational change in Car. We collected monthly vendor-level quality data for 45 months (from April 2006 to December 2009) for all suppliers of Car and Truck from Auto Inc. over multiple visits spanning more than 4 years. We do not have data from all suppliers in all 45 time periods; the data panel is thus unbalanced. Approximately

12% of the monthly records across all suppliers are missing. This occurred mainly because some suppliers did not supply in a particular month. The organizational change in Car (formation of the SIU) was implemented in January 2007; our data thus include nine preintervention months.

For this study, we excluded data from all suppliers who did not supply for at least 1 month in the pre-intervention period and at least 2 months in the post-intervention period, in order to focus exclusively on suppliers who could be expected to be affected by the implementation of the SIU. The resulting database for the 45-month period comprises 378 tier-1 suppliers to Truck and 295 tier-1 suppliers to Car. Among the 295 Car suppliers were 191 (common suppliers) who not only supplied to both Car and Truck but did so from a single location (same address, not merely the same city), and thus could be expected to experience spillover of knowledge from the Car production line to the Truck production line (and vice-versa). The period of 36 months after the organizational change gives us a good length of time to study and predict the effects on Car and Truck supply quality that followed from the deployment of the SIU in Car.

5.2 | Creation of SIU Solely Focused on Supply-Chain Quality Improvement and Addition of an Organic Structure as Treatment

The deployment of the SIU at Car fits the category of a policy change, intervention, or treatment applied to Car's sourcing (Rosenbaum 2005). It was applied to the suppliers of Car only and implemented at the plant level. The SIU was an abrupt start to a treatment for the 295 Car suppliers. Obviously, for Auto this action was not conceived as a controlled experiment, with plants chosen at random for treatment or control, but we provide robustness checks in the next section wherein we identify a sub-sample of supplier groups—treated and untreated—that are sufficiently similar as to be considered close to being randomly assigned, as well as via a synthetic control group. A similar quasi-experiment is detailed in Epplé et al. (1991). It is to be noted that our focus is on the quality of incoming components from suppliers, who are external to the car plant and do not have any influence on the internal organization of the car or truck plant. Therefore, the change in the car plant's organizational structure can be considered exogenous with respect to the analysis of incoming product quality from these external suppliers.

Since the SIU was created in Car only, and Truck's supply-chain organization remained unchanged, we consider Car to be the treated group, and Truck to be the control group for our study. Two data points can emerge in each time period, since several suppliers are common to Car and Truck. The first is for supplies to Car, and the second for supplies to Truck. Thus, the unit of analysis is buyer plant-supplier-month (e.g., Car plant-supplier A-January 2009, Truck plant-supplier A-January 2009, etc.).

5.3 | Dependent Variable

Our dependent variable $Defect_{it}$ is the natural log of defects measured in parts per million (ppm) of incoming components

from supplier i for Car and Truck in month t . Our approach is similar to that adopted in Ittner et al. (2001); the use of logarithmic scales controls for the fact that large reductions in defect rates are easier to achieve when the supplier's initial defect rates are higher than when they are already relatively low, an effect that could introduce unnecessary bias in the results. The choice of $Defect_{it}$ as our dependent variable is rooted in its critical role in assessing component quality, which is the central focus of our study. Quality improvement in this context is operationally defined as a reduction in defects. We contend that a reduction in defects is a tangible manifestation of improved quality, representing a critical measure of component quality. Our choice of Defect as the dependent variable aligns with industry practices and the real-world objectives of our research. It encompasses various aspects of component quality, such as dimensional accuracy, material defects, functional flaws, and overall compliance with specifications. Thus, it mirrors the multifaceted nature of quality in the automotive component sourcing sector.

5.4 | Moderator Variables

We deploy several variables that could be expected to account for differential improvements in quality. $Techcomp_i$ measures the technological complexity of the parts from a given supplier (which we reduce to a variable that captures the average level of technological complexity for all components supplied to Car and Truck) and thus measures the construct specified in Hypothesis 2a. We measure the complexity of supplier operations based on two factors: product complexity and manufacturing complexity. Next, we provide a brief overview of our approach to measuring complexity, but defer a more comprehensive discussion on this to Appendix F. We adopt the approach used by Novak and Eppinger (2001) and evaluate product complexity on a spectrum of 0 to 1 (low product complexity to high product complexity) as the average of three characteristics: (1) the number of subcomponents of a product, (2) product interactions, and (3) product novelty, each of which are scaled from 0 to 1. Manufacturing complexity is evaluated based on discussions with the SIU team as the average of two features: (1) the tolerance limits required for the product and (2) the sophistication involved in the manufacturing process, each of which are scaled from 0 to 1. Then, we calculate the Technical complexity of supplier operations as:

$$Techcomp_i = \frac{1}{n} \sum_{i=1}^n \frac{product_complexity_i + manufacturing_complexity_i}{2}$$

where “ n ” represents the number of distinct products sourced from each supplier. The SIU engineers computed the complexity scores for each of their respective suppliers. These scores were then validated with the vendor development and quality managers at Car independently. In our data, the complexity of supplier operations $Techcomp_i$ ranged from 0.2 (low complexity) to 1 (high complexity), with a mean of 0.69 and a standard deviation of 0.18. (Note: Complexity for Truck is similar to that for Car because of the significant overlap in sourcing. Therefore, we only use complexity calculated for Car in our analyses.).

$Supnum_{it}$ is the natural log of the number of parts supplied by a supplier to a particular plant. This variable helps to ascertain the effect of the number of parts supplied on the quality of supplied components; it measures the construct specified in Hypothesis 2b. $Supplier_locations_i$ counts the number of manufacturing locations of a supplier. This variable is used for Hypotheses 2c and 4a. $Product_commonality_i$ is used to test Hypothesis 4b. This variable represents the commonality between the components of Car and Truck for each supplier. The SIU engineers identified four factors that capture different facets of product commonality: (a) Dimensions—products with similar dimensions are expected to have higher commonality; (b) Functionality—components that play a similar role for Car's and Truck's products are expected to have higher commonality (e.g., cylinder heads); (c) Raw material—products that use similar raw material will have higher commonality; and (d) Technology—components that use similar technology will have higher commonality (e.g., pneumatically assisted gearboxes for Car will have lower commonality with the synchromesh gearboxes used for Truck). On each of these factors, four SIU engineers rated the suppliers' components on a scale between 1 and 5, where 1 indicates low commonality and 5 high commonality. The average of the ratings was taken as the measure of product commonality. $Process_commonality_i$ is used to test Hypothesis 4c. This represents the commonality in the manufacturing process of the components of Car and Truck for each supplier i and was measured by the SIU engineers as the number of process steps that are common for the manufacture of components for Car and Truck divided by the number of process steps required for the manufacture of the components for Car.

5.5 | Control Variables

Another factor that can affect quality improvement is the learning-by-doing phenomenon, which can be captured by the cumulative volume of supplied components (Lapr   et al. 2000). The notion here is that firms get better as they perform the same task repeatedly. Since our focus is on studying the differential response to Car's implementation of the SIU on quality improvement in Car and Truck, we need to control for the fact that improvements at the vendors after SIU implementation may also be driven by learning-by-doing.

Accordingly, we use cumulative volume as a control variable in our analysis: $Cumvol_{it}$ is the log of the cumulative volume of components supplied by a supplier. Table 1 in the Appendix provides means and correlations of data. Table 2 presents a summary of variables and measures.

During the observation period, we found no evidence of changes in quality assurance processes or testing rates at either Car or Truck. Both divisions maintained consistent quality assurance protocols throughout the study. This consistency was confirmed through interviews with engineers and managers at both units, as well as a review of internal documents detailing quality assurance practices.

5.6 | Empirical Research Design and Identification of Treatment Effects

We seek to determine whether, and to what extent, implementation of the SIU improved quality of incoming components for Car, and whether spillover caused changes in quality for Truck, for the same time periods. We measure the effects of implementation of the SIU in Car on the incoming quality of components by deploying difference-in-differences and triple-differences regressions on our 45-month longitudinal panel data for Car and Truck.

5.6.1 | Hypothesis 1: Difference-In-Differences

Having data sets for a treatment group (Car) and a control group (Truck) permits us to deploy the difference-in-differences (DiD) estimation. DiD is used to estimate the causal effect of a treatment or intervention by comparing the changes in outcomes over time between a population enrolled in an intervention (treatment group) and a population that is not (control group). We observe the difference in outcomes before and after the treatment for both groups and then calculate the difference between these two differences. This method helps to control for unobserved factors that could influence the outcome. By comparing the changes between the groups, we can infer the impact of the intervention, isolating it from other factors that could affect the outcome. The fundamental

TABLE 1 | Means and correlations.

Variables and Desc. Stat.				Correlations						
No	Variables	Mean	SD	1	2	2	4	5	6	7
1	Defects	6.66	2.21	1.000						
2	Techcomp	0.69	0.18	−0.0677	1.000					
3	SupNum	2.27	1.32	−0.2561	−0.161	1.000				
4	CumVol	12.78	2.05	−0.388	−0.079	0.631	1.000			
5	Supplier_Locations	3.85	3.72	−0.0123	0.331	−0.02	−0.17	1.000		
6	Product_Commonality	1.93	0.848	0.0257	−0.32	0.080	0.158	−0.481	1.000	
7	Process_Commonality	2.34	0.685	0.036	−0.191	0.023	0.1533	−0.401	0.574	1.000

TABLE 2 | Description of variables and measures.

Variable	Measure	Description
Defects	Numeric	This variable measures the quality of components sourced, in terms of parts per million defects. A lower value indicates a higher level of quality of the sourced components.
Techcomp	Numeric	This variable quantifies the level of technological complexity within the context of the auto component sourcing industry. It considers factors such as the intricacy of manufacturing processes, the degree of automation, and the level of technical expertise required. Higher values indicate higher technological complexity.
SupNum	Numeric	The number of components supplied by a given supplier is measured as a numeric count. This variable reflects the scope and scale of a supplier's engagement with the focal firm. Higher values suggest a larger portfolio of supplied components.
CumVol	Numeric	Cumulative volume represents the total volume of components supplied by a given supplier over time. It reflects the historical engagement and production output of a supplier. Higher values indicate a longer and more significant relationship with the focal firm.
Supplier_Locations	Numeric	Supplier locations refer to the geographical diversity of supplier facilities. It measures the number of different locations from which components are sourced. A higher value indicates a more dispersed supplier network.
Product_Commonality	Numeric	Product commonality assesses the extent to which different components share common features or designs. It measures the degree of similarity in the products sourced from various suppliers. Higher values indicate greater product commonality.
Process_Commonality	Numeric	Process commonality evaluates the similarity in manufacturing processes employed by different suppliers. It measures the extent to which suppliers utilize similar production methods. Higher values indicate greater process commonality.

assumption is that whatever happened to the control group (Truck) over time is what would have happened to the treatment group (Car) in the absence of the treatment (in this case, the more organic SIU structure). Unobserved time-constant heterogeneity (or omitted variable bias) is less of a problem in the DiD analysis (Card and Krueger 1994) than in multiple regression analysis.

Since we have data on multiple time periods, we base our estimates of the effects of the creation of the SIU on the incoming quality of components on DiD regressions of longitudinal panel data. In particular, we estimate the following equation (Model A):

$$Defect_{it} = Constant + b_1 \times Post_{it} + b_2 \times Treat + b_3 \times PT_{it} + Cumvol_{it} + Time_Dummies + vendor_fixed_effects + \epsilon \quad (1)$$

where $Defect_{it}$ is the dependent variable of interest for supplier i in time period t , $Post_{it}$ is a dummy variable which is 1 for time period > 9 (indicating months after the SIU creation), $Treat$ is a dummy which is 1 for the treatment group (car) and 0 for the control group (truck), and $PT_{it} = Post_{it} \times Treat$. Therefore, a negative and statistically significant value of PT_{it} would indicate significant quality knowledge generation post creation of SIU at Car. This formulation is similar to that discussed by Wooldridge (2002, 148). The coefficient of interest (the difference-in-differences policy estimator) is b_3 . $Cumvol_{it}$ is the control variable for cumulative volume.

This formulation imposes the restriction that the treatment policy has the same effect in each time period. In our study, this restriction is a basic assumption: in each successive time interval, the more organic structure of SIU drives the improvements in the same way as in other time intervals. Note that the dummy $Treat$ will get dropped in the analysis, due to fixed effects estimations. We employ monthly time dummies to control for time fixed effects in our model. Results with random effect estimations are similar.

5.6.2 | Hypotheses 2a–2c: Effect of Technological Complexity, Number of Parts, and Number of Supplier Locations

For exploring the effect of technological complexity on the difference-in-differences between Car and Truck supplies, we introduce the moderator variable $Techcomp_i$ and the interaction term of $PT_{it} \times Techcomp_i$, and call the result Model A1. Similarly, for exploring the effect of the number of parts on the difference-in-differences between Car and Truck supplies, we introduce into Model A the moderator variable $Supnum_{it}$ and the interaction term of $PT_{it} \times Supnum_{it}$, and we call the resulting Model A2. For exploring the effect of the number of manufacturing locations on the difference-in-differences between Car and Truck supplies, we introduce into Model A the moderator variable $supplier_locations_i$ and the interaction term of $PT_{it} \times supplier_locations_i$, and call the resulting Model A3. Finally, we estimate a complete model (A4) that includes all

the three interaction terms $PT_{it} \times Supnum_{it}$, $PT_{it} \times Techcomp_i$, and $PT_{it} \times supplier_locations_i$.

5.6.3 | Hypothesis 3: Difference-In-Difference-In-Differences or Triple-Differences

For identifying the presence of any spillover, we deploy the difference-in-difference-in-differences or the *triple* difference estimation, which is as follows. As defined earlier in the variable section, we estimate the difference between (a) the improvement gap between treated (Car) and untreated (never supplied to Car, control) suppliers, and (b) the improvement gap within the common suppliers supplying to both Car and Truck. The difference between these two difference-in-differences (i.e., the triple difference) tells us whether the improvement gap between all treated and untreated suppliers is higher than the improvement gap within the common suppliers. If this is so, then the smaller improvement gap in the common suppliers provides evidence of spillover in the common plants. (Note that if we focus only on common suppliers, and run a difference-in-differences analysis, then we are testing for whether there is a 100% spillover such that the improvements for both buyer plants are indistinguishable. This is a very different test from one that seeks to identify the presence of any spillover effect, which is the triple differences test that we explained above.) We estimate the following equation (Model B):

$$\begin{aligned} Defect_{it} = & Constant + b_1 \times Post_{it} + b_2 \times Group1 + b_3 \times G1_{it} \\ & + b_4 \times Group2 + b_5 \times PG2_{it} \\ & + b_6 \times PG1PG2_{it} + b_7 \times Cumvol_{it} \\ & + Time_Dummies + vendor_fixed_effects + \epsilon \end{aligned} \quad (2)$$

here, *Group1* denotes all suppliers and is coded 1 if a supplier supplies to Car and 0 otherwise. *Group2* denotes all common suppliers and is coded 1 if a *common* supplier supplies to Car and 0 if a *common* supplier supplies to Truck. Also, $PG1_{it} = Post_{it} \times Group1$, $PG2_{it} = Post_{it} \times Group2$, and $PG1PG2_{it} = Post_{it} \times Group1 \times Group2$. The coefficient of interest (the difference-in-difference-in-differences policy estimator) is b_6 . A positive and statistically significant coefficient ($p \leq 0.05$) would indicate spillover of quality knowledge. The reason we are looking for a positive sign of the triple difference interaction is because of the nature of the measurement of the spillover variables, where the defects at Truck (where spillover occurs) within common suppliers appear as negated. Therefore, a positive sign of the interaction term signifies spillover. This formulation is similar to that discussed by Wooldridge (2002, 150). Note that the dummies *Group1* and *Group2* will get dropped in the analysis due to fixed effects estimations.

Next, for delving deeper into the common suppliers, we define the dummy *Common*, which is 1 if the supplier supplies to both Car and Truck, and 0 otherwise. For these suppliers, we estimate the following equation (Model C):

$$\begin{aligned} Defect_{it} = & Constant + b_1 \times Post_{it} + b_2 \times Common + b_3 \times PC_{it} \\ & + b_4 \times Cumvol_{it} + Time_Dummies \\ & + vendor_fixed_effects + \epsilon \end{aligned} \quad (3)$$

where $PC_{it} = Post_{it} \times Common$. Note that the dummy *Common* will get dropped in the analysis due to fixed effects estimations (As noted earlier, a random effect analysis is also presented in the appendix).

5.6.4 | Hypotheses 4a–4c: Effect of Supplier Locations, Product Commonality, and Process Commonality

For exploring the effect of the number of manufacturing locations on the spillover of quality knowledge, we introduce into Model B the moderator variable *supplier_locations*, and the interaction term of $PG1PG2_i \times supplier_locations_i$, and call the resulting Model B1. Similarly, for exploring the moderation effect of product commonality on the triple difference model of spillover, Model B, we introduce the moderator variable *product_commonality*, and the interaction term of $PG1PG2_i \times product_commonality_i$, and call the result Model B2. For exploring the moderation effect of process commonality on the triple difference model of spillover, Model B, we introduce the moderator variable *process_commonality*, and the interaction term of $PG1PG2_i \times process_commonality_i$, and call the result Model B3. Finally, in model B4, we introduce all the three interactions together for a robustness check of the analysis.

6 | Results

6.1 | Test of Hypothesis 1: Effect of Addition of Organic Structure

To investigate support for Hypothesis 1, we first estimate our results using Model A for all suppliers of Car as the treated group, and all suppliers of Truck as the control group. Table 3 presents the coefficients for this estimation. For Model A, the coefficient of PT_{it} is statistically significant ($p \leq 0.001$), indicating that there is a significant difference in the improvement levels of incoming quality of components between the estimated groups of suppliers after the deployment of a more organic SIU structure in Car. This difference is after controlling for any learning effects in the suppliers to Car or Truck (Figure A1A in Appendix A). This result is consistent across the model estimates in Table 3. The main effect of coefficient of PT_{it} in A1 is -0.740 ($p < 0.001$), in A2 is -1.379 ($p < 0.001$), and in A4 is -1.249 ($p \leq 0.001$). These results support Hypothesis 1 that the addition of a more organic structure at a buyer organization is associated with quality improvement in the supply chain.

To predict the impact of the addition of a more organic structure, we follow the approach proposed by Cameron and Trivedi (2005, 103–104) for log-linear models of general form $\ln(y) = x'\beta + \epsilon$, where $E(y_i / x_i) = \exp(x_i'\beta)E[\exp(\epsilon_i)]$. Taking care of the effect of the error term, we estimate that the addition of an organic structure in the car division reduced defect rates by approximately 74.17% (calculated as $\left(\frac{\hat{y}_{PT_{it=1}} - \hat{y}_{PT_{it=0}}}{\hat{y}_{PT_{it=0}}} \right) \times 100\%$).

The DID estimator in our analysis measures the impact of the addition of a more organic organizational structure in Car, which affected the treated group (all suppliers supplying to Car experienced this change in organizational structure). The effect

TABLE 3 | Estimation results for the DID estimate of quality improvement after SIU creation (Model A, A1, A2, A3 and A4).

Dependent variable is <i>Defect_{it}</i> ; Sample: All suppliers					
Variables	Model A	Model A1	Model A2	Model A3	Model A4
<i>Post_{it}</i>	0.411*** (0.084)	0.361*** (0.064)	0.358*** (0.065)	0.357*** (0.119)	0.359*** (0.066)
<i>PT_{it}</i>	−1.272*** (0.132)	−0.740*** (0.151)	−1.379*** (0.072)	−1.715*** (0.1372)	−1.249*** (0.063)
<i>Cumvol_{it}</i>	−0.113* (0.045)	−0.098** (0.030)	−0.098** (0.030)	−0.0948*** (0.031)	−0.101** (0.032)
<i>Supnum_{it}</i>		−0.381*** (0.047)	−0.424*** (0.049)	−0.311*** (0.088)	−0.411*** (0.07)
<i>PT_{it} × Techcomp_i</i>		0.341*** (0.069)			0.326*** (0.086)
<i>PT_{it} × Supnum_{it}</i>			−0.202*** (0.045)		−0.172** (0.066)
<i>PT_{it} × supplier_locations_i</i>				0.0998*** (0.0282)	0.0828*** (0.0209)
Observations (N)	19,982	19,982	19,982	19,982	19,982
Adj. R ²	0.479	0.482	0.482	0.483	0.488

Note: Significant estimates, marked in bold, correspond to hypotheses with a significance level of at least 0.05. Cluster robust standard errors in parentheses. Includes time dummies, vendor fixed effects. Significance code: * $p < 0.05$, ** $p < 0.001$, *** $p < 0.01$.

represents the average reduction in defect rates attributed to the SIU change in Car. Our calculated DID value, approximately 74.17%, indicates that the addition of a more organic structure in the Car division led to a substantial and statistically significant reduction in defects compared to suppliers in the Truck division and signifies the improvement in product quality associated with the change in the organizational structure in the Car division. Without taking into account the effect of the errors, at the expected level, the improvement in the quality levels due to the SIU creation is 71.97%, which supports the above estimate. We refine these estimates with the inclusion of the moderating variables in the subsequent section.

The quality of components sourced by Car improved from a level of 1200 ppm on average to a level of 200 ppm on average over the period of 45 months. Truck had no significant improvement in the quality of components sourced over this period. Since there was no additional hiring, therefore, there were very little costs to Auto Inc. for deploying the organic structure in Car. While the firm has not agreed to release the cost data, yet it is reasonable to conclude that there was a net benefit.

6.2 | Test of Hypotheses 2a–2c: Effect of Technological Complexity, Number of Parts, and Number of Supplier Locations

We now explore the role of technological complexity, the number of parts supplied by suppliers, and the number of supplier locations on the difference-in-differences of quality improvement

over time between Car and Truck. In Table 3, Model A1, the coefficient of the interaction term $PT_{it} \times Techcomp_i$ is positive and significant (0.341 $p < 0.001$). In Model A2 the coefficient of the interaction term $PT_{it} \times Supnum_{it}$ is negative and significant (−0.202, $p < 0.001$). In Model A3 the coefficient of the interaction term $PT_{it} \times supplier_locations_i$ is positive and significant (0.0998, $p < 0.001$). From the combined model A4, we find that the coefficients of the three interaction terms remain significant. The parameter estimate for $PT_{it} \times Supnum_{it}$ in A4 is −0.172 ($p < 0.01$), the parameter estimate for $PT_{it} \times Techcomp_i$ is 0.326 ($p < 0.001$) and the parameter estimate for $PT_{it} \times supplier_locations_i$ is 0.0828 ($p < 0.001$). These results provide support for Hypotheses 2a–2c and indicate that

- As the number of parts supplied by a vendor increases, the difference-in-differences of quality levels between Car and Truck increases.
- As the technology complexity of a vendor increases, the difference-in-differences of quality levels between Car and Truck decreases.
- As the number of manufacturing locations of a supplier increases, the difference-in-differences of quality levels between Car and Truck decreases.

The above estimates translate to the following interpretations. The improvement in car quality for components at the mean level of technological complexity (value: 0.69) is estimated as $(1 - \exp(-0.74 + 0.341 \times 0.69) \times 100 =)39.6\%$. Similarly, the improvement of quality for components with technological complexity

at two standard deviations below the mean value (value: 0.33) of technological complexity is 46.6%. Finally, the improvement of quality for components with technological complexity at two standard deviations above the mean value (value: ≈ 1.00) of technological complexity is 32.9%. Note that the technological complexity of 1 is also the maximum value for the variable. Therefore, we find that the percentage improvement reduces as the value of technological complexity increases, thus supporting Hypothesis 3. Furthermore, we find that quality improvement over the entire range of the technological complexity measure is significant, with a minimum level of quality improvement of approximately 33% even when the value of technological complexity is at its maximum. These estimates further nuance the effect of organizational change on supplier quality as expressed in Hypotheses 1 and 2a.

The improvement in quality of components for suppliers supplying only 1 component is estimated to be 79.4%. However, this improvement increases to 83.1% for suppliers supplying 2 components, 86.8% for suppliers supplying 3 components, and 90.8% for suppliers supplying 5 components. These estimates support the Hypothesis 2b.

Finally, the improvement in quality of components for suppliers with 1 manufacturing location is 80.12% in Car as compared to that in Truck. However, this improvement reduces to 78.03% for suppliers with two manufacturing locations, 75.72% for suppliers with three manufacturing locations, 73.17% for suppliers with four

manufacturing locations, 70.36% for suppliers with four manufacturing locations, and 67.25% for suppliers with six manufacturing locations. These estimates support Hypothesis 2c and nuance the results related to Hypotheses 1 and 2c.

6.3 | Test of Hypothesis 3: Spillover of Quality Knowledge

In Table 4 we present the results of Model B and Model C corresponding to the spillover of quality knowledge. For Model B, the triple-difference coefficient of $PG1PG2_{it}$ is not significant, indicating that there is no statistically significant difference between the differences in the improvement levels of incoming quality of components of (a) treated and untreated suppliers, and (b) common suppliers (the variable $PG2_{it}$ is dropped, being collinear with the triple difference estimate $PG1PG2_{it}$). We, therefore, postulate that a significant transfer of learning did not take place in the common supplier factories, even though these supplier factories were supplying similar components to both plants. The suppliers did not improve their quality for all their buyer plants. Car received significantly better quality components, even from the common suppliers (Figure A1B in Appendix A).

For the third and the fourth (Model C) estimations, the coefficient of PC_{it} is not statistically significant, indicating that there is no significant difference in the improvement levels of

TABLE 4 | Estimation results for the triple difference estimate of quality knowledge spillover.

Dependent variable is <i>Defect_{it}</i> ; Sample: All suppliers for model B, B1, B2, B3, B4; respective suppliers for model C							
Variables	Model B	Model C (Car)	Model C (Truck)	Model B1	Model B2	Model B3	Model B4
<i>Post_{it}</i>	0.405*** (0.063)	0.146** (0.071)	−0.174 (0.119)	0.572*** (0.151)	0.565*** (0.151)	0.563*** (0.152)	0.573*** (0.151)
<i>PG1_{it}</i>	−1.064*** (0.281)			−1.715*** (0.209)	−0.625*** (0.151)	−0.619** (0.184)	−2.169*** (0.601)
<i>PG1PG2_{it}</i>	−0.237 (0.305)			−0.212 (0.313)	−0.315 (0.326)	−0.332 (0.311)	−0.324 (0.303)
<i>PC_{it}</i>		−0.189 (0.305)	−0.051 (0.124)				
<i>Cumvol_{it}</i>	−0.112* (0.045)	−0.571** (0.082)	−0.154** (0.047)	−0.294*** (0.0708)	−0.291*** (0.071)	−0.289*** (0.072)	−0.295*** (0.071)
<i>PG1PG2_{it} × supplier_locations_i</i>				−0.0998** (0.0282)			−0.1072** (0.034)
<i>PG1PG2_{it} × Product_commonality_i</i>					0.357** (0.123)		0.426** (0.140)
<i>PG1PG2_{it} × Process_commonality_i</i>						0.299** (0.091)	0.533* (0.240)
Observations (N)	19,982	9224	10,758	19,982	19,982	19,982	19,982
Adj. R ²	0.478	0.393	0.498	0.484	0.481	0.482	0.486

Note: Cluster robust standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Includes time dummies, and vendor fixed effects.

incoming quality of components between the common and the not-common suppliers of Car or Truck after the deployment of Car's SIU. Common supplier exposure to the SIU initiatives did not affect the quality improvements in components supplied to Truck. Our data and analysis suggest that the benefits of investment by the buyer do not necessarily spill over at the supplier end to other buyers, even when the other buyer is another unit of the same firm (Figure A2 in Appendix A). These results provide strong support for Hypothesis 1, but do not support Hypothesis 3.

6.4 | Test of Hypotheses 4a–4c: Factors Associated With Significant Spillover

While from Model B and Model C we find that, in general, the spillover of quality knowledge is not significant as a main effect, we find from Model B1 that the spillover of quality knowledge, post creation of SIU, is significantly associated with the number of locations of manufacturing units of a supplier. Many of the suppliers in the sample have multi-location manufacturing. This indicates that the chance of spillover of quality knowledge is relatively higher and significant for colocated manufacturing. This finding is significant because if buyers are concerned about quality knowledge spillover, then they can select suppliers with distributed manufacturing. This result supports Hypothesis 4a. Similarly, we find that the interactions of product and process commonality with quality knowledge spillover are significant and positive. This supports Hypotheses 4b and 4c. This means that product and process commonality significantly increase the chances of spillover.

In the Auto industry and many other high-tech industries, often competitors share the same or similar parts and components. For example, many auto firms share the same brakes, air bags, transmissions, and even engines. In such cases, buyers need to be aware of the chances of quality knowledge spillover when working with the supply chain to generate quality knowledge (Agrawal et al. 2016). In sum, while we do not find support for significant spillover for all suppliers, under the following conditions, spillover of quality knowledge can occur statistically significantly:

1. Co-located manufacturing of different buyers' parts and components.
2. High product commonality such as same and similar components.
3. High process commonality in the manufacturing of parts and components of different buyers.

7 | Robustness Checks

In this section, we explain several tests that we deployed to check the robustness of our results. The detailed methods and supporting results are in Appendix D.

7.1 | Matching and Synthetic Controls

In our study, the suppliers to Car (treatment group) had been chosen by Car, not at random; therefore, our estimates of the impact of Car's SIU implementation may be biased by the existence of

systematic differences between suppliers to Car and those who do not supply to Car. We use propensity score matching to check the robustness of the estimation of treatment effects, controlling for the existence of these confounding differences (Rosenbaum and Rubin 1983). The propensity score matching design helps us control for differences in covariates between the groups, so that we can attribute differences in the quality of the sourced components from the two groups to the deployment of Car's SIU (treatment). We repeated the differences-in-differences analysis for our subset of data and found that there is a significant difference in the improvement levels of incoming quality of components between the propensity score-matched subset of the two groups of suppliers, after the change of structure in Car. Table E1 in Appendix D, Columns 2 and 3 (Models A and A3) show the coefficients for this estimation for matched suppliers for quality improvement. We also repeated the triple differences analysis for our subset of data to check the results related to spillover. Table E2 in Appendix D, Columns 2 and 3 (Models B and B4) show the coefficients for this estimation for matched suppliers for quality improvement spillover. The robustness checks support our main results.

Next, even though the quality intervention does not happen for Truck, our analysis assumes this difference affects the mean quality of sourced components between the two divisions. Yet, one can suggest that Truck is still not an appropriate control group for analysis simply due to the nature of differences between the two divisions during and after the treatment period. We address this issue using the synthetic control method (SCM), which is used to identify the treatment effect when an untreated group does not provide an appropriate comparison (Abadie et al. 2010) and is extended to identify the treatment effect when the treatment group includes multiple exposed subjects (Cavallo et al. 2013), such as multiple suppliers in our case. Table E1 in Appendix D, Columns 4 and 5 (Models A and A3) show the coefficients for this estimation for SCM matched suppliers for quality improvement. Table E2 in Appendix D, Columns 4 and 5 (Models B and B4) show the coefficients for this estimation for SCM matched suppliers for quality improvement spillover. We find that our insights in our main models remain the same with this approach, suggesting that Truck provides an appropriate comparison for the treatment group of Car. We also observe that the PSM and SCM methods both provide a slightly higher estimate for the policy change effect compared to the main model. From a conservative perspective, we use the main model estimates for the managerial discussion.

Next, as a precursor to the difference-in-differences analysis, we performed a simple univariate analysis on defects to identify whether the data have a basic variation with respect to the treatment. This check shows that Car and Truck have parallel trends before the treatment (month 9); however, after the treatment, the average defects decrease for Car but not for Truck. We also deploy a different test for examining the parallel trends assumption of the difference-in-differences models as in Gallino and Moreno (2014), and use

$$Y_{it} = \text{Constant} + b_1 \text{Treat}_i \times \text{Trend}_{it} + b_2 \text{Trend}_{it} + \text{Controls}_{it} b_C + \text{vendor_fixed_effects} + \epsilon \quad (4)$$

where Trend_{it} counts the number of months since the beginning of the dataset. We run Equation (4) using pre-intervention data.

A statistically significant interaction coefficient would indicate that the pre-intervention trends of the two supplier groups are not parallel, violating the fundamental parallel trends assumption. In all cases, we find no significance for the interaction terms, indicating that the parallel trends assumption holds for our data. Fourth, during the 45 months for which the data is available, new models were introduced in both Car and Truck. Since a new model often requires a new component, providing suppliers with both a challenge (stabilizing quality for a new production line) and an opportunity (a product or process design change is easier to introduce, achieving discrete improvement in quality), we needed to consider the impact of model changes on quality levels. However, the volumes of these new models were not significant in the observation period. Moreover, measures of the finished quality of Auto Inc. products (e.g., external surveys, such as those of JD Power) indicated no major change in build quality. It thus seemed unlikely that new model introduction had a significant effect on our study. However, we tested the data for evidence of the effects of new models by including indicator variables for new models to control for model changes in all our analyses and found that our results and inferences remain robust to these additional controls.

Next, we conducted a placebo test by keeping only observations before the SIU intervention (i.e., observations until month 9) and defined month 5 as the fake policy change date. We do not find any statistically significant effect on our variable of interest due to this fictitious policy change.

Finally, we also analyzed the data using random-effect models. Table E3 in Appendix D presents the random effect analyses for our data along with comments. Random effect analysis results are completely aligned with the fixed effect analysis results. We do not use random effect formulations as our main analyses because the Hausman test indicates that fixed-effect formulations are appropriate for our data.

7.2 | Exploring Alternative Explanations and Confounding Factors

In this section, we explore alternative explanations for our results and discuss a couple of confounding factors. First, the analysis of Model A does not preempt the alternative explanation that Truck is probably different from Car in some fundamental way that could explain a difference in outcomes between the two plants. If this explanation is correct, then we should find a difference between the quality improvements for those suppliers who supply to Truck versus those suppliers who do not supply to Truck. Table 4, column 3 (Model C for Car) shows this analysis. We see that the coefficient of the difference-in-differences estimator PC_{it} is not significant. Therefore, we can assert that on average, there is no significant difference-in-differences of quality improvement between the suppliers to Truck (the common suppliers) and other suppliers of Car (the not-common suppliers): These suppliers do not differ in any fundamental way.

A second alternative explanation could be that the treated (Car) buyer and the suppliers devoted more resources to quality than the non-treatment buyer (Truck). We explored this alternative explanation at length with Car executives. We verified that,

before SIU, Car and Truck were devoting roughly equal resources to monitoring the quality of inputs from these suppliers. It was not the case that Truck was, all along, more lax about devoting resources to quality control than Car. Nor were additional resources added, either by suppliers or by either division of Auto Inc., when the SIU was created; Car simply shifted the focus of a part of its existing quality resources—it directed those resources to stop working on short-term quality control in-house (such as the separation of bad parts received on the buyer end) and start working on long-term quality improvement at the supplier end (such as arresting defects at source).

A third possibility we considered is that perhaps Car had chosen to source better quality parts, either by signaling that quality was important (and the suppliers fell in line), or by weeding out bad-quality suppliers. Our data and empirical work with suppliers and managers of Auto Inc. do not support this explanation. We talked with suppliers at length about this issue of communication and signaling, and our conclusion is this: the suppliers were willing to spend time and money, and when they could sense that the changes suggested were useful, they responded. The SIU engineers slowly built up data suggesting why defects were occurring and how these defects could be arrested. We posit that it was neither a threatening atmosphere nor a simple signaling process that produced the changes; rather, it was the creation of a non-threatening, focused-learning environment by the deployment of SIU's focused role and more organic structure that resulted in improvements.

Finally, we address the possibility that the observed quality improvements in Car could be attributed to a gradual improvement in quality culture rather than the structural changes brought about by the SIU formation. Quality culture is a critical driver of product quality improvement, influencing organizational actions through shared values, beliefs, and routines (Cameron and Sine 1999; Detert et al. 2000; Brahm and Poblete 2024; Maull et al. 2001; Hackman and Wageman 1995). Defined as the “general approach, values, and orientation toward quality that permeate organizational actions” (Cameron and Sine 1999), quality culture changes can motivate actors to focus on continuous improvement. However, these changes are incremental and require sustained leadership, communication, and employee engagement (Baird et al. 2011; Cannon et al. 2010; Sousa and Voss 2001).

The literature consistently highlights that quality culture change is gradual and challenging. Shifting from a focus on error detection to error prevention can significantly improve quality outcomes, yet “changes in organizational culture are slow and difficult” (Cameron and Sine 1999; Denison 1990). Deeply ingrained norms and resistance to change necessitate ongoing communication, trust-building, and alignment of individual and organizational goals. This process demands consistent reinforcement and support, often over extended periods (Atkinson 1993; Handfield and Ghosh 1994).

Staudenmayer et al. (2002) emphasize that organizational change is further hindered by operational pressures. Employees responsible for production and delivery often prioritize immediate needs over long-term improvement actions, focusing on error detection “rather than error prevention.” Structural shifts,

however, can alleviate these pressures by providing the necessary time and resources for exploratory activities. For instance, in their study of three organizations, Staudenmayer et al. (2002) observed that breaks from routine temporal constraints allowed employees to engage in long-term improvement efforts. They noted that “people suddenly found themselves cut loose from regular temporal routines, and thus were more open to change” (p. 590).

In the context of Car, while quality culture change might have contributed to quality improvement, the immediate and discontinuous improvements observed suggest that structural changes, such as the creation of the SIU, played a more pivotal role by enabling employees to focus on long-term quality improvement. This underscores the complementary yet distinct roles of structural shifts and cultural evolution in driving organizational change.

To test this alternate explanation, we employ Regression Discontinuity Design (RDD), which is particularly suited for identifying causal impacts associated with discrete structural changes. This method allows us to isolate immediate and discontinuous shifts in quality performance linked to the SIU creation, thereby distinguishing structural impacts from slower, culture-driven changes.

RDD is a quasi-experimental approach that uses local regression techniques to estimate the discontinuous change in an outcome variable resulting from a threshold or cutoff in an independent variable, called the forcing variable (Hahn et al. 2001; Calonico et al. 2014). This method is well suited for analyzing structural changes caused by discrete events, such as policy changes or organizational restructuring (Yilmaz et al. 2024; Park et al. 2020; Calvo et al. 2019; Cong and Howell 2021). Specifically, the RDD model we estimate has the following structure as in Equation (5).

$$\arg \min_{\beta_0, D, \beta_1, \beta_2, \gamma} \sum_{i=1}^n \sum_{t=0}^T K\left(\frac{t-t_0}{h}\right) \left[\log \text{Defects}_{it} - \left(\beta_0 + D \mathbb{I}(t \geq t_0) + \beta_1 (t-t_0)^- + \beta_2 (t-t_0)^+ + \gamma^T \mathbf{X}_i \right) \right]^2 \quad (5)$$

where, $K(\cdot)$ is a kernel function, h is the bandwidth, t is time, t_0 is the time of event (SIU creation), β_0 is the intercept, D is the discontinuity estimate at $t = t_0$ which is the parameter of interest, β_1 is the local slope at $t < t_0$, β_2 is the slope at $t \geq t_1$, and γ is the vector of slopes for covariates $\mathbf{X}$. A negative and significant estimate of D indicates that at the time t_0 there has been a sharp drop in defect rates of the supplied components. This would indicate that the creation of SIU had a sharp and near-immediate effect on the quality of supplied components, an effect that is not likely to be achieved through a general change in the quality culture of “Car”. We used the same set of covariates as the main DID model. Also, we used the Gaussian kernel for the first-order local polynomial estimates. The event time for the RDD estimate is coded as $t_0 = 0$, that is, $t < 0$ indicates pre-SIU and $t \geq 0$ indicates post-SIU periods.

RDD graphs in Figure H1 show both the linear and quadratic effects of time for Car and Truck. We observe that there has been a sharp decrease in the defect rates of supplied components for Car. However, from the estimates of Truck, we see that there has been no such decrease in defect rates of supplied components. This indicates that the creation of SIU led to an immediate improvement in the quality of supplied components for Car.

The RDD regression estimates for both Car and Truck in Table H1 show that the estimate for the discontinuity parameter “D” in Equation (5) is -0.6864 with an estimated p -value of 1.98×10^{-5} for Car. This indicates that the SIU creation helped reduce the defect rate significantly for Car. This estimate is equivalent to a defect rate reduction to 23.58 PPM (95% CI: [15.38, 36.15]) in the short term after the creation of the SIU, from a defect rate of 46.84 PPM (95% CI: [35.11, 62.50]) before the creation of the SIU. The slope of time (the rate at which the defect rate reduces per week) is estimated to be -0.0815 with a p -value of 0.002. This indicates that the rate at which the defect rate reduces is also significant. After 5 weeks of the creation of SIU the estimated defect rate reduces to 10.44 PPM (95% CI: [9.52, 25.84]), indicating sustenance of efforts by the SIU in improving the quality of sourced components at the origin of the problem, that is, error prevention rather than error detection and rejection. The estimates of RDD for Truck are neither significant for the discontinuity nor significant for the rate of reduction of defects over time. The discontinuity estimate for Truck at the time the SIU was formed is -0.1549 with a p -value of 0.173. These estimates, established in a quasi-experimental causal regression using RDD, indicate that the results were indeed driven by the change in organizational structure in Car rather than a general improvement in the quality culture in Car.

A confounding factor that we explored was the role of supplier operators and engineers. During our visits, the supplier managers, without exception, acknowledged the contribution of their SIU engineer in improving their quality. The supplier operators and engineers were involved in this improvement. Almost all our interviews echoed a consistent theme—many of the issues related to quality problems are complex and often require knowledge of processes and design that is not residing at the

supplier side alone. Without active buyer involvement, it is not possible to reduce the ambiguity around what is the root cause of the problem. The problem is—this search for the root cause (know-why) takes time, and the change in structure by Car provided this opportunity. A second confounding question was: what factors prevented Car from making this change earlier? Why make the change now? In our interviews, we did not find any special reasons for the specific timing of the SIU intervention. The Car plant manager, however, revealed that the decision to implement this change was made because he had concluded that long-term quality improvement at suppliers was an increasingly important priority, and that such improvement could not be achieved within the (mechanistic) structure then deployed in Car’s sourcing.

8 | Conclusion

Organizational design consists of multiple interrelated elements that shape firm performance. Two critical aspects are the incentive systems that align individual motivation with

long-term versus short-term goals, and the structural framework that governs how decision rights are distributed. A focus on long-term quality improvement requires engineers to engage in deeper problem-solving and a more nuanced understanding of processes. This, in turn, demands both appropriate incentives and an enabling structure. An organic and flatter structure, by decentralizing authority and enhancing autonomy, promotes exploration and allows engineers to pursue sustained quality improvements rather than merely respond to immediate operational issues. We bring novel data to a problem that has been hard to research quantitatively, by capitalizing on a unique opportunity in a natural quasi-experimental setup. In our field study, the same firm changed its supplier-related incentives through changes in the roles of a part of supply chain quality engineers, enabled by a more organic organization structure and practices in one plant but not the other, and made available to us data for both plants on quality, before and after the change. Our extensive fieldwork, longitudinal data, analyses of these data, and theoretical explanations suggest that the changes operationalized by Car transformed its sourcing. In Auto Inc., the deployment of a more organic structure enabled organizational agents to focus on long-term quality improvement goals and engage in a better understanding of cause and effect, and led to better outcomes because the learning was more specific. It is useful to note that the SIU change, from the perspective of the car plant, is not exogenous. However, since we analyze the quality of components generated by organizations external to the car or truck plant, and since the external suppliers do not have any influence on the internal organization of the car or truck, we consider the change as exogenous with respect to the analysis of the quality of the incoming product.

8.1 | Organizational Incentives, Structure and Supply Chain Quality

When firms devote resources to quality improvement, quality can be expected to improve. Our study extends existing understanding of approaches to quality improvement by showing, with longitudinal data, how the creation of a dedicated team with incentives focused solely on supply-chain quality improvement, combined with a shift to a more organic structure at a buyer, results in improved quality outcomes at suppliers. Both incentives generated through focused roles and task allocation, and a supporting organic structure for greater independence and authority of decision making down the line are key elements of an exploration enabling organizational design. We are not claiming that the quality improvement methods we observed are novel; rather, we posit that the effect of a structural change at a buyer on quality improvement at its suppliers and related contingencies has not previously been studied. The important element of this study is the organizational structure link of the SIU. It is natural to ponder whether the quality improvement in Car could have been deployed in the absence of the reporting relationship of the SIU. Our detailed interviews helped us tease out this effect better. The creation of a team focused on quality improvement can help provide a pathway to reducing ambiguity in quality improvement. However, if team members are still working within a social scenario where everyone else might continue to behave in the same manner, including their manager, this would provide mixed signals for team

members to engage in the desired (long-term) goals. For the SIU in car, the change in structure provided the additional lever of changing this social environment and providing extrinsic motivation for long-term goals. This could happen since the SIU was associated with the strategic branch of the organizational structure (i.e., the Car plant manager) and the engineers were more likely to engage in long-term strategic actions since their external reward structure was more likely to be aligned with strategic actions (Wood and Bandura 1989). In our fieldwork involving 19 SIU engineers and 102 suppliers, we consistently observed that the structural separation provided by the SIU was instrumental in overcoming production pressures and facilitating long-term quality improvements. The SIU engineers were exclusively tasked with quality enhancement, with little to no operational distractions. Suppliers also reported a shift from reactive quality management to proactive engagement driven by SIU engineers, which they attributed to the changed and focused roles and structural autonomy of the SIU. Overall, our long-term work with Auto Inc. leads us to suggest that without organizing differently, it would be much more difficult for an organization to make long-term changes in its sourcing. Finally, we do not claim that the changes implemented at “Car” are the only way to achieve improvements in supplier quality. Other approaches may also be effective, such as switching suppliers or embedding quality expectations in contracts with financial penalties for poor incoming quality. However, we suggest that the combination of role-based incentives, achieved through focused task allocation, and a supporting organic structure that enables greater independence and decision-making authority down the line, represents a validated approach. This combination offers a credible, empirically supported way for buyer organizations to influence the quality of incoming parts and components from their supply chains.

8.2 | Contributions

Our study empirically examines the effect of focal buyer's changes in incentives through the redesign of roles and a shift in organizational structure from a relatively more mechanistic to a relatively more organic structure on supply chain quality and knowledge creation. We contribute by demonstrating that organizations can change internal incentives and organizational structure to influence how organizational agents interact with external supply chain partners to create long-term quality knowledge and improve long-term quality performance. To our knowledge, ours is the first paper that demonstrates how the internal organizational design of a buyer can influence supply chain partners and create value. We contribute to the knowledge creation literature (e.g., Lapré et al. 2000) by providing evidence that know-why at the suppliers can be created by providing long-term incentives at the buyer. Organizational redesign at a buyer can help deploy a new performance management system, which can help create long-term incentives at the buyer's end. We also contribute to the interface between organization design, comprising both role design and resulting changes in implicit incentives, and associated changes in an enabling organizational structure, and supply chain literature (e.g., Sosa et al. 2004) by providing evidence that improved quality outcomes may not emerge naturally over time in the absence of such changes. We also

contribute to the knowledge spillover literature (e.g., Argote et al. 2000) by demonstrating that the effects of the change of the structure at the buyer can be compartmentalized such that even other plants in the buyer firm, not to mention plants owned by competitors, may not benefit from improvements realized by that buyer. Supplier knowledge gained from one relationship with a buyer firm does not automatically spill over to other relationships, even with other divisions of the same buyer firm.

Our findings are especially relevant to operating managers, as they provide new light on the management of quality improvement at suppliers and can enable more informed decision making. First, we find that changes in organizational design can change the implicit incentive structure of organizational agents and help focus on specific goals and activities. Second, OEMs are often concerned about knowledge spillovers in supply chains while investing in knowledge generation. We alleviate such concerns by demonstrating that quality knowledge generated within supply chains is sticky and does not easily spill over, possibly due to product and process specificity, co-location, and supply-side organizational barriers to knowledge spillover. Identification of these effects can help focal buyers prioritize quality knowledge-related investments in supply chains.

8.3 | Generalization of Findings

Our study is empirically motivated by a quasi-experiment at a specific organization, Auto Inc., in a specific industry. Although the specificities may not be generalizable to other contexts, such as other firms or industries, the theoretical development helps us to delineate generalizable insights from the specific context. First, our finding that changes in implicit incentives and organizational structure of a focal buyer can influence and enable long-term performance improvements in the supply chain is generalizable and adds new evidence-based theoretical insight to the literature. The creation of a dedicated team focused on improving supplier quality generates implicit non-financial incentives for greater exploration of new quality-related knowledge and enables independent decision-making through an organic structure. This finding is generalizable to other similar contexts. Second, the imperative of creating specific organizational changes and mechanisms, and the directional causal relationship with long-term knowledge creation, can be generalizable from this study. Finally, the finding that the specificity of product and process knowledge, and geographic dispersion of manufacturing, inhibits spillovers of created knowledge, is generalizable and can act as a theoretical pointer toward prioritizing engagements of focal buyers on improvement activities within their supply chains.

We examine the quality performance of a large set of vendors supplying two plants of a single large firm, yet our findings may be specific to the automotive industry. It remains to be seen whether similar effects of organizational redesign are observed in other industries. Second, we identify the joint effect of two interdependent mechanisms: structural, through a shift toward a relatively more organic structure, and incentives, through differentiation and focus on goal formation, that intervene in quality improvement. However, our study does not isolate the

relative importance of these mechanisms, as they operate complementarily in our context. The quasiexperimental nature of our study limits our ability to isolate the individual effects of the two key changes: the incentive shift driven by the redesign of SIU engineers' roles and the transition to a more organic organizational structure. These changes were closely interlinked; the shift toward a long-term quality improvement role required a structural change that allowed greater exploration, while the structural change alone would likely have had a limited impact without the corresponding role redefinition. Despite this interdependence, we believe that it is meaningful to consider the potential differential impact of each dimension on the quality improvement of supplied parts and components. We acknowledge this as a limitation of our study, as both changes occurred simultaneously. Future research could explore their independent effects. Finally, we acknowledge that the effects we observe are associative in nature and while our study tests one mechanism, organizational restructuring, future studies could investigate the comparative effectiveness of other mechanisms. We hope that our work stimulates further theoretical and empirical research on the role of organizational design in improving supply chain quality. In closing, we believe that we took advantage of a unique opportunity in the field to test a mechanism that is useful to managers to improve supply chain quality and has immense practical value for networked and outsourced organizations.

We are indebted to the managers at Auto Inc. for helping us explore many supply chain concepts. We thank (late) Prof. Huseyin Leblebici and Prof. Joseph Mahoney from the University of Illinois Urbana-Champaign (UIUC) for their extremely helpful comments on this paper. This work was partially supported by a grant from the Campus Research Board, UIU.

References

- Abadie, A., A. Diamond, and J. Hainmueller. 2010. "Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California's Tobacco Control Program." *Journal of the American Statistical Association* 105, no. 490: 493–505.
- Adler, P. S., and B. Borys. 1996. "Two Types of Bureaucracy: Enabling and Coercive." *Administrative Science Quarterly* 41, no. 1: 61–89.
- Aghion, P., and J. Tirole. 1997. "Formal and Real Authority in Organizations." *Journal of Political Economy* 105, no. 1: 1–29.
- Agrawal, A., Y. Kim, H. D. Kwon, and S. Muthulingam. 2016. "Investment in Shared Suppliers: Effect of Learning, Spillover, and Competition." *Production and Operations Management* 25, no. 4: 736–750.
- Agrawal, A., U. Mukherjee, and S. Muthulingam. 2020. "Does Organizational Forgetting Affect Quality Knowledge Gained Through Spillover?—Evidence From the Automotive Industry." *Production and Operations Management* 29, no. 4: 907–934.
- Alcácer, J., and W. Chung. 2007. "Location Strategies and Knowledge Spillovers." *Management Science* 53, no. 5: 760–776.
- Argote, L., P. Ingram, J. M. Levine, and R. L. Moreland. 2000. "Knowledge Transfer in Organizations: Learning From the Experience of Others." *Organizational Behavior and Human Decision Processes* 82, no. 1: 1–8.
- Argote, L., and E. Miron-Spektor. 2011. "Organizational Learning: From Experience to Knowledge." *Organization Science* 22, no. 5: 1123–1137.

- Atkinson, P. E. 1993. "Creating Culture Change: The Key to Successful Total Quality Management." *Journal of the Operational Research Society* 44: 525.
- Baird, K., K. J. Hu, and R. Reeve. 2011. "The Relationships Between Organizational Culture, Total Quality Management Practices and Operational Performance." *International Journal of Operations & Production Management* 31, no. 7: 789–814.
- Balachandran, K. R., and S. Radhakrishnan. 2005. "Quality Implications of Warranties in a Supply Chain." *Management Science* 51, no. 8: 1266–1277.
- Bardhan, I., V. V. Krishnan, and S. Lin. 2013. "Team Dispersion, Information Technology, and Project Performance." *Production and Operations Management* 22, no. 6: 1478–1493.
- Barney, J. B. 1985. "Information Cost and the Governance of Economic Transactions." In *Organizations and Markets*, edited by R. D. Nacmali and A. Rugiadini, 347–372. Societa Editrice it Milano.
- Bensaou, B., and E. Anderson. 1999. "Buyer-Supplier Relations in Industrial Markets: When Do Buyers Risk Making Idiosyncratic Investments?" *Organization Science* 10, no. 4: 460–481.
- Bondareva, M., and E. Pinker. 2019. "Dynamic Relational Contracts for Quality Enforcement in Supply Chains." *Management Science* 65, no. 3: 1305–1321.
- Brahm, F., and J. Poblete. 2024. "Organizational Culture, Adaptation, and Performance." *Organization Science* 35: 1823–1848.
- Burns, T., and G. M. Stalker. 1994. *The Management of Innovation*. Oxford University Press.
- Burton, R. M., B. Erisen, D. D. Hakonsson, and C. C. Snow. 2006. *Organization Design: The Evolving State-Of-The-Art*, 19–40. Springer.
- Burton, R. M., and B. Obel. 2018. "The Science of Organizational Design: Fit Between Structure and Coordination." *Journal of Organization Design* 7, no. 1: 1–13.
- Cadsby, C. B., F. Song, and F. Tapon. 2007. "Sorting and Incentive Effects of Pay for Performance: An Experimental Investigation." *Academy of Management Journal* 50, no. 2: 387–405.
- Calonico, S., M. D. Cattaneo, and R. Titiunik. 2014. "Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs." *Econometrica* 82, no. 6: 2295–2326.
- Calvo, E., R. Cui, and J. C. Serpa. 2019. "Oversight and Efficiency in Public Projects: A Regression Discontinuity Analysis." *Management Science* 65, no. 12: 5651–5675.
- Cameron, A. C., and P. K. Trivedi. 2005. *Microeconometrics: Methods and Applications*. Cambridge University Press.
- Cameron, K., and W. Sine. 1999. "A Framework for Organizational Quality Culture." *Quality Management Journal* 6, no. 4: 7–25.
- Cannon, J. P., P. M. Doney, M. R. Mullen, and K. J. Petersen. 2010. "Building Long-Term Orientation in Buyer-Supplier Relationships: The Moderating Role of Culture." *Journal of Operations Management* 28, no. 6: 506–521.
- Cao, H., X. Guan, T. Fan, and L. Zhou. 2020. "The Acquisition of Quality Information in a Supply Chain With Voluntary vs. Mandatory Disclosure." *Production and Operations Management* 29, no. 3: 595–616.
- Card, D., and A. B. Krueger. 1994. "Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania." *American Economic Review* 84, no. 4: 772–793.
- Cardinale, I. 2018. "Beyond Constraining and Enabling: Toward New Microfoundations for Institutional Theory." *Academy of Management Review* 43, no. 1: 132–155.
- Cavallo, E., S. Galiani, I. Noy, and J. Pantano. 2013. "Catastrophic Natural Disasters and Economic Growth." *Review of Economics and Statistics* 95, no. 5: 1549–1561.
- Chao, G. H., S. M. R. Iravani, and R. C. Savaskan. 2009. "Quality Improvement Incentives and Product Recall Cost Sharing Contracts." *Management Science* 55, no. 7: 1122–1138.
- Chen, L., S. Yao, and K. Zhu. 2020. "Responsible Sourcing Under Supplier-Auditor Collusion." *Manufacturing & Service Operations Management* 22, no. 6: 1234–1250.
- Chen, Y.-J., S. Seshadri, and E. Zemel. 2008. "Sourcing Through Auctions and Audits." *Production and Operations Management* 17, no. 2: 121–138.
- Child, J. 1973. "Predicting and Understanding Organization Structure." *Administrative Science Quarterly* 18: 168–185.
- Clark, K. B., and T. Fujimoto. 1991. *Product Development Performance*. HBS Press.
- Cohen, W. M., and D. A. Levinthal. 1990. "Absorptive Capacity: A New Perspective on Learning and Innovation." *Administrative Science Quarterly* 35, no. 1: 128–152.
- Cong, L. W., and S. T. Howell. 2021. "Policy Uncertainty and Innovation: Evidence From Initial Public Offering Interventions in China." *Management Science* 67, no. 11: 7238–7261.
- Constas, H. 1958. "Max Weber's Two Conceptions of Bureaucracy." *American Journal of Sociology* 63, no. 4: 400–409.
- Crosby, P. B. 1967. *Cutting the Cost of Quality*. Industrial Education Institute.
- Csaszar, F. A. 2013. "An Efficient Frontier in Organization Design: Organizational Structure as a Determinant of Exploration and Exploitation." *Organization Science* 24, no. 4: 1083–1101.
- Cyert, R. M., and J. G. March. 1959. "A Behavioral Theory of Organizational Objectives." In *Modern Organization Theory*, 76–90. Prentice-Hall.
- da Silveira, G. J. C., B. Snider, and J. Balakrishnan. 2013. "Compensation-Based Incentives, Erp and Delivery Performance: Analysis From Production and Improvement Perspectives." *International Journal of Operations & Production Management* 33, no. 4: 415–441.
- Daft, R. L., and R. H. Lengel. 1986. "Organization Information Requirements, Media Richness, and Structural Design." *Management Science* 32, no. 5: 554–571.
- Dai, Y., S. X. Zhou, and Y. Xu. 2012. "Competitive and Collaborative Quality and Warranty Management in Supply Chains." *Production and Operations Management* 21, no. 1: 129–144.
- d'Aspremont, C., and A. Jacquemin. 1988. "Cooperative and Noncooperative R & D in Duopoly With Spillovers." *American Economic Review* 78, no. 5: 1133–1137.
- Darr, E. D., L. Argote, and D. Epple. 1995. "The Acquisition, Transfer, and Depreciation of Knowledge in Service Organizations: Productivity in Franchises." *Management Science* 41, no. 11: 1750–1762.
- Darr, E. D., and T. R. Kurtzberg. 2000. "An Investigation of Partner Similarity Dimensions on Knowledge Transfer." *Organizational Behavior and Human Decision Processes* 82, no. 1: 28–44.
- De Leeuw, S., and J. P. Van Den Berg. 2011. "Improving Operational Performance by Influencing Shopfloor Behavior via Performance Management Practices." *Journal of Operations Management* 29, no. 3: 224–235.
- de Zegher, J. F., D. A. Iancu, and H. L. Lee. 2019. "Designing Contracts and Sourcing Channels to Create Shared Value." *Manufacturing & Service Operations Management* 21, no. 2: 271–289.
- Dehejia, R. H., and S. Wahba. 2002. "Propensity Score-Matching Methods for Nonexperimental Causal Studies." *Review of Economics and Statistics* 84, no. 1: 151–161.
- Deming, W. E. 1986. *Out of the Crisis*. MIT Press.

- Denison, D. 1990. "Corporate Culture and Organizational." New York: Wiley.
- Dike, P.(2013). The Impact of Workplace Diversity on Organizations. Dobbin, F., & Jung, J.(2010). Corporate Board Gender Diversity and Stock Performance: The Competence Gap or Institutional Investor Bias." *NCL Rev* 89: 809.
- Detert, J. R., R. G. Schroeder, and J. J. Mauriel. 2000. "A Framework for Linking Culture and Improvement Initiatives in Organizations." *Academy of Management Review* 25, no. 4: 850–863.
- Dhanorkar, S. S., E. Siemsen, and K. W. Linderman. 2018. "Promoting Change From the Outside: Directing Managerial Attention in the Implementation of Environmental Improvements." *Management Science* 64, no. 6: 2535–2556.
- Donaldson, L. 2006. "The Contingency Theory of Organizational Design: Challenges and Opportunities." In *Organization Design: The Evolving State-Of-The-Art*, 19–40. Springer.
- Dong, B., W. Tang, and C. Zhou. 2020. "Strategic Sourcing Under Recall Loss Sharing and Product Quality Investment." *RAIRO-Operations Research* 54, no. 4: 1133–1160.
- Dong, Y., K. Xu, Y. Xu, and X. Wan. 2016. "Quality Management in Multi-Level Supply Chains With Outsourced Manufacturing." *Production and Operations Management* 25, no. 2: 290–305.
- Dougherty, D. 2001. "Reimagining the Differentiation and Integration of Work for Sustained Product Innovation." *Organization Science* 12, no. 5: 612–631.
- Douglas, T. J., and W. Q. Judge. 2001. "Total Quality Management Implementation and Competitive Advantage: The Role of Structural Control and Exploration." *Academy of Management Journal* 44, no. 1: 158–169.
- Dyer, J. H. 1996. "Specialized Supplier Networks as a Source of Competitive Advantage: Evidence From the Auto Industry." *Strategic Management Journal* 17, no. 4: 271–291.
- Dyer, J. H. 1997. "Effective Interfirm Collaboration: How Firms Minimize Transaction Costs and Maximize Transaction Value." *Strategic Management Journal* 18, no. 7: 535–556.
- Epplé, D., L. Argote, and R. Devadas. 1991. "Organizational Learning Curves: A Method for Investigating Intra-Planet Transfer of Knowledge Acquired Through Learning by Doing." *Organization Science* 2, no. 1: 58–70.
- Fisher, M., K. Ramdas, and K. Ulrich. 1999. "Component Sharing in the Management of Product Variety: A Study of Automotive Braking Systems." *Management Science* 45, no. 3: 297–315.
- Fynes, B., and C. Voss. 2001. "A Path Analytic Model of Quality Practices, Quality Performance, and Business Performance." *Production and Operations Management* 10, no. 4: 494–513.
- Galbraith, J. R. 1974. "Organization Design: An Information Processing View." *Interfaces* 4, no. 3: 28–36.
- Gallino, S., and A. Moreno. 2014. "Integration of Online and Offline Channels in Retail: The Impact of Sharing Reliable Inventory Availability Information." *Management Science* 60, no. 6: 1434–1451.
- Gambardella, A., C. Panico, and G. Valentini. 2015. "Strategic Incentives to Human Capital." *Strategic Management Journal* 36, no. 1: 37–52.
- Garvin, G. A. 1984. "What Does Product Quality Really." *Sloan Management Rev* 1: 1984.
- Gittelman, M. 2007. "Does Geography Matter for Science-Based Firms? Epistemic Communities and the Geography of Research and Patenting in Biotechnology." *Organization Science* 18, no. 4: 724–741.
- Gokpinar, B., W. J. Hopp, and S. M. R. Iravani. 2010. "The Impact of Misalignment of Organizational Structure and Product Architecture on Quality in Complex Product Development." *Management Science* 56, no. 3: 468–484.
- Gray, J. V., A. V. Roth, and M. J. Leiblein. 2011. "Quality Risk in Offshore Manufacturing: Evidence From the Pharmaceutical Industry." *Journal of Operations Management* 29, no. 7–8: 737–752.
- Greenwood, R., and D. Miller. 2010. "Tackling Design Anew: Getting Back to the Heart of Organizational Theory." *Academy of Management Perspectives* 24, no. 4: 78–88.
- Gualandris, J., R. D. Klassen, S. Vachon, and M. Kalchschmidt. 2015. "Sustainable Evaluation and Verification in Supply Chains: Aligning and Leveraging Accountability to Stakeholders." *Journal of Operations Management* 38: 1–13.
- Ha, A. Y., W. Shang, and Y. Wang. 2023. "Supplier Audit Information Sharing and Responsible Sourcing." *Management Science* 69, no. 1: 308–324.
- Hackman, J. R., and R. Wageman. 1995. "Total Quality Management: Empirical, Conceptual, and Practical Issues." *Administrative Science Quarterly* 40: 309–342.
- Hahn, J., P. Todd, and W. Van der Klaauw. 2001. "Identification and Estimation of Treatment Effects With a Regression-Discontinuity Design." *Econometrica* 69, no. 1: 201–209.
- Handfield, R., and S. Ghosh. 1994. "Creating a Quality Culture Through Organizational Change: A Case Analysis." *Journal of International Marketing* 2, no. 3: 7–36.
- Handley, S. M., and J. V. Gray. 2013. "Inter-Organizational Quality Management: The Use of Contractual Incentives and Monitoring Mechanisms With Outsourced Manufacturing." *Production and Operations Management* 22, no. 6: 1540–1556.
- Harris, M., and A. Raviv. 2002. "Organization Design." *Management Science* 48, no. 7: 852–865.
- Helper, S., and M. Sako. 1995. "Supplier Relations in Japan and the United States: Are They Converging." *Sloan Management Review* 36, no. 3: 77–85.
- Huber, G. P., and R. R. McDaniel. 1986. "The Decision-Making Paradigm of Organizational Design." *Management Science* 32, no. 5: 572–589.
- Ishikawa, K. 1985. *Total Quality Control*. Prentice-Hall Inc.
- Ittner, C. D., V. Nagar, and M. V. Rajan. 2001. "An Empirical Examination of Dynamic Quality-Based Learning Models." *Management Science* 47, no. 4: 563–578.
- Jacobs, M. A., and M. Swink. 2011. "Product Portfolio Architectural Complexity and Operational Performance: Incorporating the Roles of Learning and Fixed Assets." *Journal of Operations Management* 29, no. 7–8: 677–691.
- Jaffe, A. B., M. Trajtenberg, and R. Henderson. 1993. "Geographic Localization of Knowledge Spillovers as Evidenced by Patent Citations." *Quarterly Journal of Economics* 108, no. 3: 577–598.
- Jensen, M. C., and W. H. Meckling. 2009. "Specific and General Knowledge, and Organizational Structure." In *Knowledge Management and Organisational Design*, 17–38. Routledge.
- Joseph, J., R. Klingebiel, and A. J. Wilson. 2016. "Organizational Structure and Performance Feedback: Centralization, Aspirations, and Termination Decisions." *Organization Science* 27, no. 5: 1065–1083.
- Juran, J. M. 1951. *Quality Control Handbook*. McGraw-Hill.
- Kang, M. P., J. T. Mahoney, and D. Tan. 2009. "Why Firms Make Unilateral Investments Specific to Other Firms: The Case of OEM Suppliers." *Strategic Management Journal* 30, no. 2: 117–135.
- Karpoﬀ, J. M. 2001. "Public Versus Private Initiative in Arctic Exploration: The Effects of Incentives and Organizational Structure." *Journal of Political Economy* 109, no. 1: 38–78.
- Keidel, R. W. 1994. "Rethinking Organizational Design." *Academy of Management Perspectives* 8, no. 4: 12–28.

- Klassen, R. D., and S. Vachon. 2003. "Collaboration and Evaluation in the Supply Chain: The Impact on Plant-Level Environmental Investment." *Production and Operations Management* 12, no. 3: 336–352.
- Kretschmer, T., and P. Puranam. 2008. "Integration Through Incentives Within Differentiated Organizations." *Organization Science* 19, no. 6: 860–875.
- Lambert, R. A., D. F. Larcker, and K. Weigelt. 1993. "The Structure of Organizational Incentives." *Administrative Science Quarterly* 38: 438–461.
- Lampel, J., and C. Giachetti. 2013. "International Diversification of Manufacturing Operations: Performance Implications and Moderating Forces." *Journal of Operations Management* 31, no. 4: 213–227.
- Lane, P. J., and M. Lubatkin. 1998. "Absorptive Capacity and Interorganizational Learning." *Strategic Management Journal* 19, no. 5: 461–477.
- Lapr , M. A., A. S. Mukherjee, and L. V. Wassenhove. 2000. "Behind the Learning Curve: Linking Learning Activities to Waste Reduction." *Management Science* 46, no. 5: 597–611.
- Lascelles, D. M., and B. G. Dale. 1989. "The Buyer-Supplier Relationship in Total Quality Management." *Journal of Purchasing and Materials Management* 25, no. 2: 10–19.
- Lavie, D., U. Stettner, and M. L. Tushman. 2010. "Exploration and Exploitation Within and Across Organizations." *Academy of Management Annals* 4, no. 1: 109–155.
- Lee, H.-H., and C. Li. 2018. "Supplier Quality Management: Investment, Inspection, and Incentives." *Production and Operations Management* 27, no. 2: 304–322.
- Lee, S., and P. Meyer-Doyle. 2017. "How Performance Incentives Shape Individual Exploration and Exploitation: Evidence From Microdata." *Organization Science* 28, no. 1: 19–38.
- Leifer, R., and G. P. Huber. 1977. "Relations Among Perceived Environmental Uncertainty, Organization Structure, and Boundary-Spanning Behavior." *Administrative Science Quarterly* 22: 235–247.
- Levine, D. I., and M. W. Toffel. 2010. "Quality Management and Job Quality: How the Iso 9001 Standard for Quality Management Systems Affects Employees and Employers." *Management Science* 56, no. 6: 978–996.
- Li, Z., S. Q. Lu, J. K. Ryan, and D. Sun. 2021. "Impact of Organizational Structure on Development Strategy Under Equity-Based Incentives." *Production and Operations Management* 30, no. 4: 984–996.
- Lippman, S. A., and R. P. Rumelt. 1982. "Uncertain Imitability: An Analysis of Interfirm Differences in Efficiency Under Competition." *Bell Journal of Economics* 13, no. 2: 418–438.
- Lu, T., and B. Tomlin. 2022. "Sourcing From a Self-Reporting Supplier: Strategic Communication of Social Responsibility in a Supply Chain." *Manufacturing & Service Operations Management* 24, no. 2: 902–920.
- March, J. G., and H. A. Simon. 1958. *Organizations*. John Wiley & Sons.
- Marengo, L., and C. Pasquali. 2012. "How to Get What You Want When You Do Not Know What You Want: A Model of Incentives, Organizational Structure, and Learning." *Organization Science* 23, no. 5: 1298–1310.
- Maull, R., P. Brown, and R. Cliffe. 2001. "Organisational Culture and Quality Improvement." *International Journal of Operations & Production Management* 21, no. 3: 302–326.
- Milliken, F. J. 1990. "Perceiving and Interpreting Environmental Change: An Examination of College Administrators' Interpretation of Changing Demographics." *Academy of Management Journal* 33, no. 1: 42–63.
- Mintzberg, H. 1989. *The Structuring of Organizations*. Springer.
- Mishra, A., K. K. Sinha, S. Thirumalai, and A. Van de Ven. 2020. "Sourcing Structures and the Execution Efficiency of Information Technology Projects: A Comparative Evaluation Using Stochastic Frontier Analysis." *Journal of Operations Management* 66, no. 3: 281–309.
- Modi, S. B., and V. A. Mabert. 2007. "Supplier Development: Improving Supplier Performance Through Knowledge Transfer." *Journal of Operations Management* 25, no. 1: 42–64.
- Mukherjee, A. S., M. A. Lapre, and L. N. Van Wassenhove. 1998. "Knowledge Driven Quality Improvement." *Management Science* 44, no. 11-part-2: S35–S49.
- Muthulingam, S., and A. Agrawal. 2016. "Does Quality Knowledge Spillover at Shared Suppliers? An Empirical Investigation." *Manufacturing & Service Operations Management* 18, no. 4: 525–544.
- Nadler, D., M. Tushman, and M. B. Nadler. 1997. *Competing by Design: The Power of Organizational Architecture*. Oxford University Press.
- Natter, M., A. Mild, M. Feurstein, G. Dorffner, and A. Taudes. 2001. "The Effect of Incentive Schemes and Organizational Arrangements on the New Product Development Process." *Management Science* 47, no. 8: 1029–1045.
- Nishiguchi, T. 1994. *Strategic Industrial Sourcing*. Oxford University Press.
- Nonaka, I. 1994. "A Dynamic Theory of Organizational Knowledge Creation." *Organization Science* 5, no. 1: 14–37.
- Novak, S., and S. D. Eppinger. 2001. "Sourcing by Design: Product Complexity and the Supply Chain." *Management Science* 47, no. 1: 189–204.
- Oliva, R., and J. D. Sterman. 2001. "Cutting Corners and Working Overtime: Quality Erosion in the Service Industry." *Management Science* 47, no. 7: 894–914.
- Park, S., E. Rabinovich, C. S. Tang, and R. Yin. 2020. "The Impact of Disclosing Inventory-Scarcity Messages on Sales in Online Retailing." *Journal of Operations Management* 66, no. 5: 534–552.
- Pierce, L. 2012. "Organizational Structure and the Limits of Knowledge Sharing: Incentive Conflict and Agency in Car Leasing." *Management Science* 58, no. 6: 1106–1121.
- Pleshko, L., and I. Nickerson. 2008. "Strategic Orientation, Organizational Structure, and the Associated Effects on Performance in Industrial Firms." *Academy of Strategic Management Journal* 7, no. 1: 95–110.
- Posen, H. E., and J. S. Chen. 2013. "An Advantage of Newness: Vicarious Learning Despite Limited Absorptive Capacity." *Organization Science* 24, no. 6: 1701–1716.
- Pugh, D. S., D. J. Hickson, and C. R. Hinings. 1969. "An Empirical Taxonomy of Structures of Work Organizations." *Administrative Science Quarterly* 14, no. 1: 115–126.
- Ramasubbu, N., J. Shang, J. H. May, Y. Tjader, and L. Vargas. 2019. "Task Interdependence and Firm Performance in Outsourced Service Operations." *Manufacturing & Service Operations Management* 21, no. 3: 658–673.
- Roberts, J. 2007. *The Modern Firm: Organizational Design for Performance and Growth*. Oxford University Press.
- Rosenbaum, P. R. 2005. "Observational Study." *Encyclopedia of Statistics in Behavioral Science* 3: 1451–1462.
- Rosenbaum, P. R., and D. B. Rubin. 1983. "The Central Role of the Propensity Score in Observational Studies for Causal Effects." *Biometrika* 70, no. 1: 41–55.
- Roth, A., J. Singhal, K. Singhal, and C. S. Tang. 2016. "Knowledge Creation and Dissemination in Operations and Supply Chain Management." *Production and Operations Management* 25, no. 9: 1473–1488.
- Rowan, B. 1982. "Organizational Structure and the Institutional Environment: The Case of Public Schools." *Administrative Science Quarterly* 27: 259–279.

- Sager, F., and C. Rosser. 2009. "Weber, Wilson, and Hegel: Theories of Modern Bureaucracy." *Public Administration Review* 69, no. 6: 1136–1147.
- Sine, W. D., H. Mitsuhashi, and D. A. Kirsch. 2006. "Revisiting Burns and Stalker: Formal Structure and New Venture Performance in Emerging Economic Sectors." *Academy of Management Journal* 49, no. 1: 121–132.
- Sosa, M. E., S. D. Eppinger, and C. M. Rowles. 2004. "The Misalignment of Product Architecture and Organizational Structure in Complex Product Development." *Management Science* 50, no. 12: 1674–1689.
- Sosa, M. E., J. Mihm, and T. R. Browning. 2013. "Linking Cyclicalities and Product Quality." *Manufacturing & Service Operations Management* 15, no. 3: 473–491.
- Sousa, R., and C. A. Voss. 2001. "Quality Management: Universal or Context Dependent?" *Production and Operations Management* 10, no. 4: 383–404.
- Sroufe, R., and S. Curkovic. 2008. "An Examination of Iso 9000: 2000 and Supply Chain Quality Assurance." *Journal of Operations Management* 26, no. 4: 503–520.
- Staats, B. R., D. J. Brunner, and D. M. Upton. 2011. "Lean Principles, Learning, and Knowledge Work: Evidence From a Software Services Provider." *Journal of Operations Management* 29, no. 5: 376–390.
- Staudenmayer, N., M. Tyre, and L. Perlow. 2002. "Time to Change: Temporal Shifts as Enablers of Organizational Change." *Organization Science* 13, no. 5: 583–597.
- Steinbach, A. L., T. R. Holcomb, R. M. Holmes Jr., C. E. Devers, and A. A. Cannella Jr. 2017. "Top Management Team Incentive Heterogeneity, Strategic Investment Behavior, and Performance: A Contingency Theory of Incentive Alignment." *Strategic Management Journal* 38, no. 8: 1701–1720.
- Stiglitz, J. E. 1989. "Incentives, Information, and Organizational Design." *Empirica* 16, no. 1: 3–29.
- Szulanski, G. 1996. "Exploring Internal Stickiness: Impediments to the Transfer of Best Practice Within the Firm." *Strategic Management Journal* 17, no. S2: 27–43.
- Thompson, J. D. 1967. *Organizations in Action*. McGraw Hill.
- Thornton, R. A., and P. Thompson. 2001. "Learning From Experience and Learning From Others: An Exploration of Learning and Spillovers in Wartime Shipbuilding." *American Economic Review* 91, no. 5: 1350–1368.
- Trent, R. J., and R. M. Monczka. 1999. "Achieving World-Class Supplier Quality." *Total Quality Management* 10, no. 6: 927–938.
- Tushman, M. L., and D. A. Nadler. 1978. "Information Processing as an Integrating Concept in Organizational Design." *Academy of Management Review* 3, no. 3: 613–624.
- Westphal, J. D., and E. J. Zajac. 2013. "A Behavioral Theory of Corporate Governance: Explicating the Mechanisms of Socially Situated and Socially Constituted Agency." *Academy of Management Annals* 7, no. 1: 607–661.
- Wiseman, R. M., and L. R. Gomez-Mejia. 1998. "A Behavioral Agency Model of Managerial Risk Taking." *Academy of Management Review* 23, no. 1: 133–153.
- Wood, R., and A. Bandura. 1989. "Social Cognitive Theory of Organizational Management." *Academy of Management Review* 14, no. 3: 361–384.
- Wooldridge, J. M. 2002. *Econometric Analysis of Cross Section and Panel Data*. MIT Press.
- Wright, P., M. Kroll, J. A. Krug, and M. Pettus. 2007. "Influences of Top Management Team Incentives on Firm Risk Taking." *Strategic Management Journal* 28, no. 1: 81–89.
- Yeung, A. C. L. 2008. "Strategic Supply Management, Quality Initiatives, and Organizational Performance." *Journal of Operations Management* 26, no. 4: 490–502.
- Yilmaz, Ö., Y. Son, G. Shang, and H. A. Arslan. 2024. "Causal Inference Under Selection on Observables in Operations Management Research: Matching Methods and Synthetic Controls." *Journal of Operations Management* 70, no. 5: 831–859.
- Zander, U., and B. Kogut. 1995. "Knowledge and the Speed of the Transfer and Imitation of Organizational Capabilities: An Empirical Test." *Organization Science* 6, no. 1: 76–92.
- Zhang, D., K. Linderman, and R. G. Schroeder. 2012. "The Moderating Role of Contextual Factors on Quality Management Practices." *Journal of Operations Management* 30, no. 1–2: 12–23.
- Zhang, Q., and X. Sun. 2023. "How Incentive Synergy and Organizational Structures Shape Innovation Ambidexterity." *Journal of Knowledge Management* 27, no. 1: 156–177.

Appendix A

Graphs for Car Versus Truck, and Common Suppliers

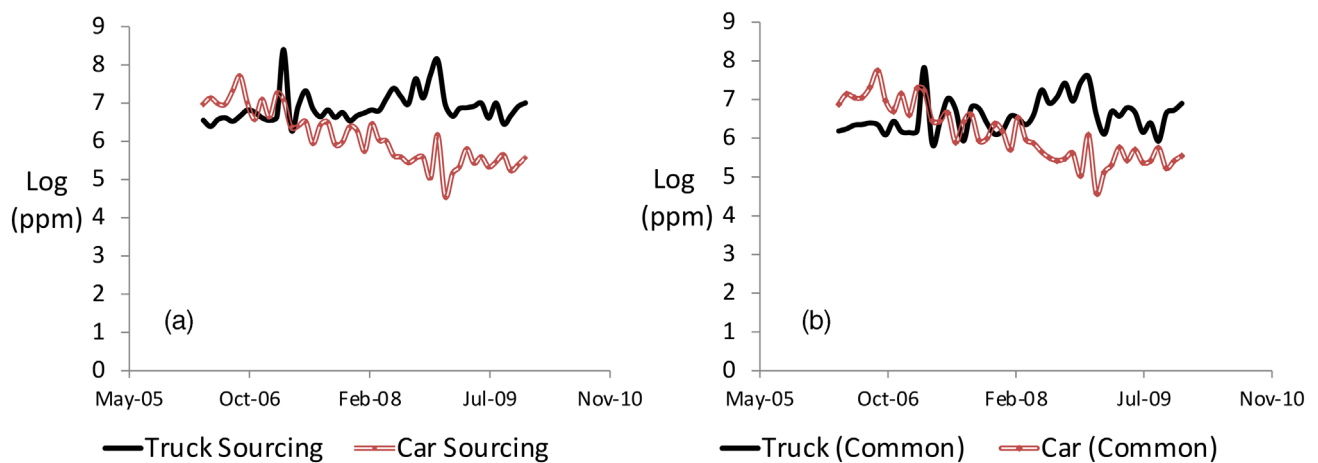


FIGURE A1 | (A) Quality trend for all suppliers of the car and the truck divisions. (B) Quality trend for common suppliers of the car and the truck divisions. The y axis has log of defects, so a lower point on the graph indicates better quality.

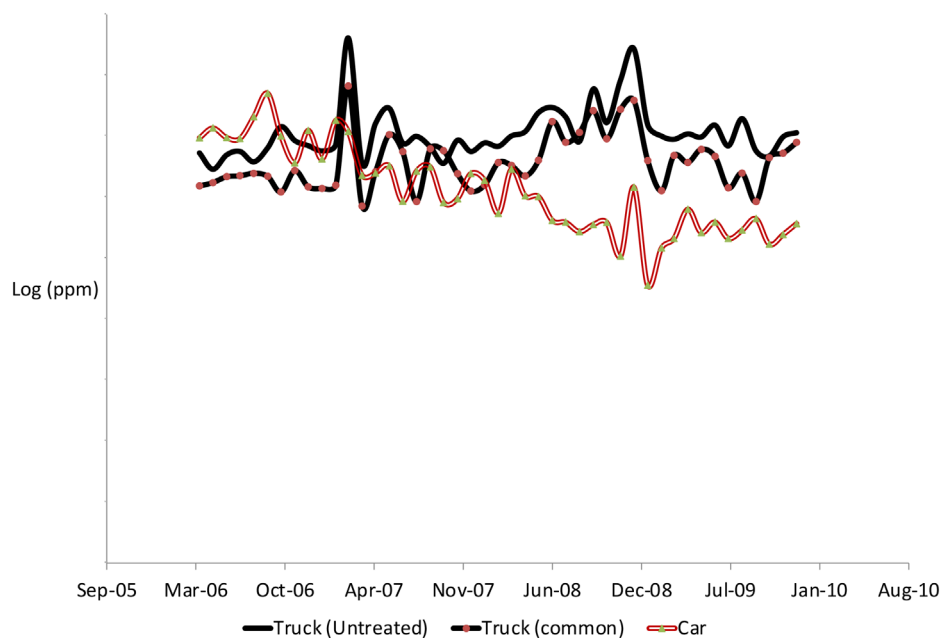


FIGURE A2 | Quality trend for common and not common truck suppliers, and car suppliers.

Appendix B

Examples of Quality Improvements Deployed by the SIU

TABLE B1 | Problems, root cause identification, improvements, time taken in months (rounded).

No	Type of improvement	Problem	Root cause/diagnosis	Improvement	Time
1	Acceptance criteria improvement	Uneven chamfer in Semi Trailing arm LH	Incorrect QA specs, welding issue	(1) Chamfer specs redefined with Engineering, min 1.5 mm, drawing changed (2) Welding fixture corrected to ensure correct chamfer after milling operation	3
2	Layout modified	Inlet valve rejections in assembly	Excessive material handling, manual control	Separate manufacturing line established with new machine sequence.	11
3	New Inspection method	Input shaft fouling with shaft cover	Burr in trimming operation	Inspection gauge for 21 mm hole. Gauge locating at 35.3 dia hole. Deburring included in process sheet, Drawing modified.	5
4	Operator/Inspector training	Rubber hose wall thickness variation	Mounting process out of control	Operators trained to (i) peel off extra rubber layer during mounting, (ii) do consistent wrapping to minimize overlap	7
5	Packing/Logistics improvement	Adjustment screw threads of Parking brake lever uneven	Screw thread getting damaged in transit	Protective rubber sleeve introduced to prevent thread damage due to handling	2
6	Specification Correction	Assembly of reinforcement qtr difficult	Striker plate mismatch	Panel reinforcement rear quarter corrected as per drawing, Hole location shifted as per CAD data; New master sample made	4
7	Pokayoke	Assembly bracket fitment and assembly issue	3 mm spring retention hole position shifted downwards due to reverse orientation of lever in drilling fixture	Pokayoke done by incorporating stoppers in drilling fixture (prevents fitment in reverse orientation)	1
8	Preventive maintenance	Sealing face of Rear Cover leaking. Dimension required 3 mm, observed 2.4 mm.	Excess wear out of embossing pad observed resulting in tool breakdown	Breakdown maintenance of tooling defined. New emboss pad made as spare.	4
9	Process Correction	Nugget fail defect in Assembly frame front door RH	Weak weld spot between Cross member longitudinal front, Lower arm & MBR front crash RH	(i) Provided packing at clamps—RH Side to reduce gap (ii) Shift 5 nos. spot positions by 8 mm at front side.	7
10	Process improvement	Variation in starting notches of parking brake lever	Starting point of notch varies during milling operation	V block introduced to locate the base ratchet while milling and eliminate variation of 1st notch.	4
11	Product design change	Breakage on Assembly bracket (shifting) pin threading	After carburising stress relieving of threads is done through flame softening.	Drawing modified to induction hardening of working surface to have pin with soft threads	14
12	Sub Supplier Improvement	Heavy chatter and dent marks on machining surface	Supplies from tier-2 supplier highly variable	Machining supplier changed from XX to YY. Process improved at YY.	10
13	Tooling Improvement	10 Dia hole on Engine B Mount lower shifted	Component location shifts during piercing operation	Holder block bolts replaced by dowells. Operator retrained.	4

Appendix C

Detailing an Example of Quality Improvement

It is instructive to detail an improvement initiative (Case 10 in Table B1) undertaken at a Car supplier of Auto Inc. which can help explain the learning related to root causes involved for the supplier (know-why) as well as the degree of buyer-supplier interaction that was necessary for effective knowledge generation. In this case, the milling of components (Parking Brake Lever) performed at the supplier had variations in the starting point of the first notch of the lever, even for components within the same production lot. A careful study by the Car SIU engineer and supplier revealed that ten components were being machined at once, and all of these were not resting in exactly the same alignment against the flat support. The solution involved replacing the flat support with a “V” block so

that component-level vibrations could be avoided (see right panel in Figures A2 and C1).

Our discussions with the SIU engineer highlighted that even after experimentation and improvement, it took time to internalize the reason for the usage of the “V” block, including developing the knowledge that correct alignment of all ten components in one step is essential. If components are not fully clamped in a single step, it may lead to misaligned components, which may cause further variations in the milling operation. Moreover, things change over time: the “V” block can develop surface irregularities due to repeated use, and this may lead to uneven placement of components, causing variations in the milled surfaces. Thus, if a new operator is deployed who does not clamp the components correctly or does not perform adequate preventive maintenance of the “V” block, then milling defects can recur.

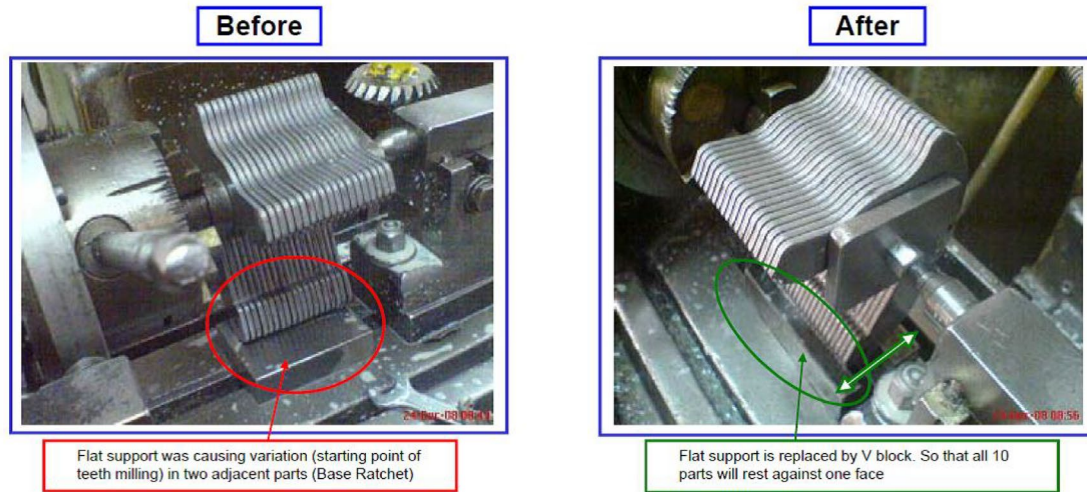


FIGURE C1 | Example of quality improvement initiative at car's supplier.

Appendix D

Empirical Framework

See Figure D1.

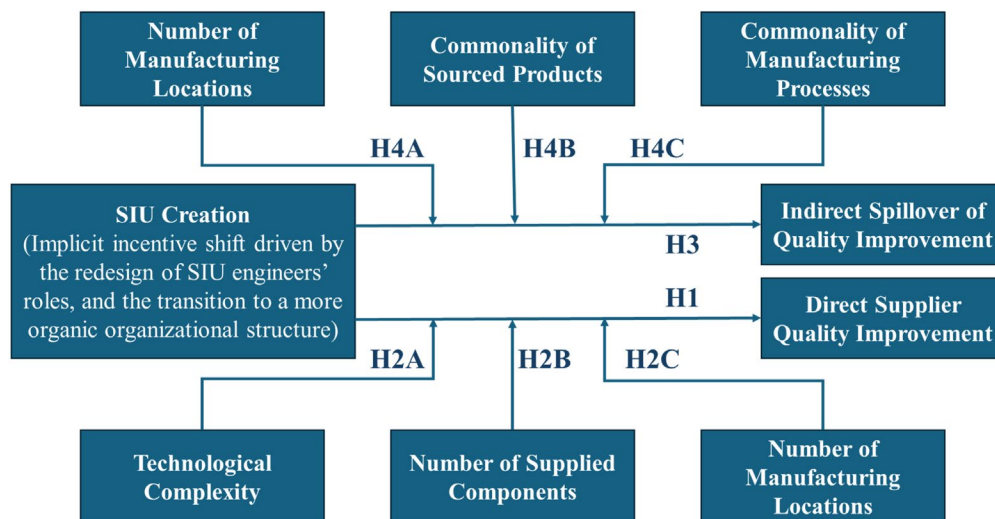


FIGURE D1 | Research framework for empirical research.

Appendix E

Robustness Tests

Propensity Score Matching

In our study, the suppliers to Car (treatment group) had been chosen by Car, not at random; therefore, our estimates of the impact of Car's SIU implementation may be biased by the existence of systematic differences between suppliers to Car and those who do not supply to Car. We use propensity score matching to check the robustness of the estimation of treatment effects, controlling for the existence of these confounding differences (Rosenbaum and Rubin 1983). The propensity score matching design helps us control for differences in covariates between the groups so that we can attribute differences in the quality of the sourced components from the two groups to the deployment of Car's SIU (treatment).

To estimate these similar groups, we proceeded as follows. We selected the 10th month as a sample month of treatment and used the quality levels (log of defects) in periods 1 to 9 as covariates, in addition to other covariates such as technology complexity, supplier volume, and the number of parts supplied. We estimated the propensity score for each supplier by using a logistic regression model in which treatment status was regressed on the covariates. Following Dehejia and Wahba (2002), we utilized radius matching without replacement, employing the common support restriction with a narrow caliper range (0.01). We tested the reduction in bias via radius matching to the reduction in bias using one-to-one matching and found that radius matching was better. The average bias in quality covariates of unmatched groups was 24%, and in other covariates (technology complexity, volume) was 25%. We achieved an average reduction of 71% in bias for all covariates by matching; *t*-tests for differences between means of treated and control groups for the matched sample were insignificant in all cases. Thus, matching on the propensity score resulted in a control group whose distribution of covariates was similar to that of the treated group (Figure E1).

Synthetic Control Method

The premise of the Synthetic Control Method is that a synthetic control group obtained through counterfactuals from a combination of control group subjects can provide a better comparison for the treatment group than any single subject in the control group alone. Unlike in propensity score matching in which similar subjects are identified using only observable covariates, in SCM we identify a control group using not only observable covariates but also pre-intervention outcomes; thus the synthetic control is developed as the weighted average of all potential control group subjects that best resemble the characteristics of the treatment group subjects. Hence, a synthetic control group generated with

this method in our study can be considered a hypothetical control group that (1) includes suppliers that were supplying to Car during both the pre-intervention period and the post-intervention period and (2) is obtained through counterfactuals using the support of data. Tables E1–E3, F1, H1, Columns 3 and 4 provide the difference-in-differences model results obtained from the comparison of the treatment group to the synthetic control group.

We estimate the random effects difference-in-differences model for three subgroups of data. The first estimation (Model A) uses all suppliers of the car division as the treated group and all suppliers of the truck division as the control group. Model A is as follows:

$$\begin{aligned} Defect_{it} = & Constant + b_1 \times Post + b_2 \times Treat + b_3 \times PT \\ & + b_4 \times Supnum_{it} + b_5 \times Techcomp_{it} + b_6 \times Cumvol_{it} \\ & + Time_Dummies + error \end{aligned}$$

where $PT = Post \times Treat$.

The second estimation (Model A1) is aimed at identifying the difference in *common* suppliers—these suppliers are supplying components to both divisions from the same geographical location (same manufacturing factory). This estimation is for a reduced number of suppliers (*common*) and uses Model A, with *common*.

The third estimation (Model B) is for the DID of quality improvement for the truck division between its common and uncommon suppliers. This estimation tells us how suppliers supplying to the truck division were affected by the participation in the car program. The *common* suppliers supplied to the car division, and therefore they were—being at the same geographical location—exposed to the improvements and initiatives of the car division. The uncommon suppliers have no relationship whatsoever with the car division. Model B is as follows

$$\begin{aligned} Defect_{it} = & Constant + b_1 \times Post + b_2 \times Common + b_3 \times PC \\ & + b_4 \times Supnum_{it} + b_5 \times Techcomp_{it} + b_6 \times Cumvol_{it} \\ & + Time_Dummies + error \end{aligned}$$

where $PC = Post \times Common$.

The fourth estimation (Model B1) is for the DID of quality improvement for the car division between its common and uncommon suppliers.

For the first two estimations, the DID coefficient *PT* is highly significant, indicating that there is a significant difference in the

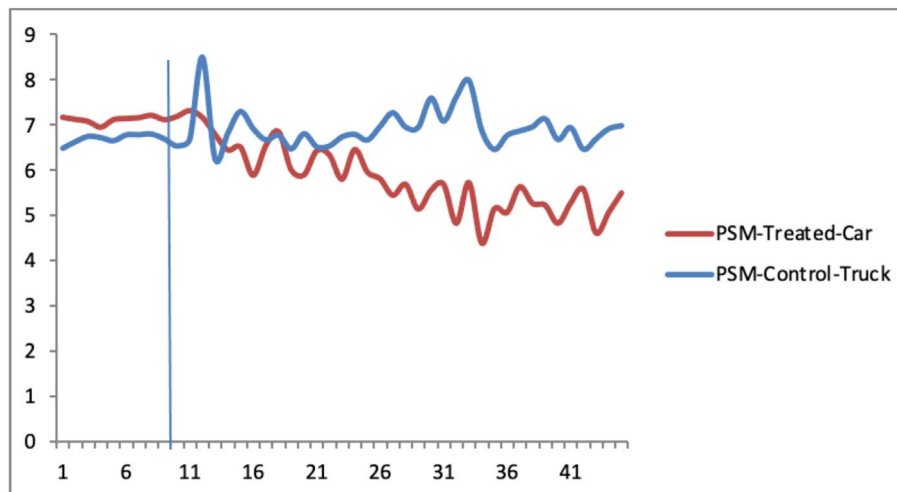


FIGURE E1 | Quality trend for propensity score matched car and truck suppliers.

TABLE E1 | Robustness checks for quality improvement using propensity matching and Synthetic control.

Dependent variable is <i>Defect_{it}</i>				
Variables	Matched suppliers		Synthetic control	
Models	A	A3	A	A3
<i>Post_{it}</i>	0.324*** (0.090)	0.272*** (0.073)	0.409*** (0.078)	0.361*** (0.065)
PT_{it}	−1.201*** (0.096)	−1.172*** (0.071)	−1.312*** (0.101)	−1.227*** (0.098)
<i>Cumvol_{it}</i>	−0.125** (0.042)	−0.118** (0.039)	−0.197*** (0.029)	−0.189*** (0.033)
<i>Supnum_{it}</i>	−0.371 (0.065)	−0.379** (0.121)	−0.278** (0.097)	−0.286** (0.090)
<i>PT_{it} × Supnum_{it}</i>		−0.203** (0.071)		−0.222** (0.072)
<i>PT_{it} × Techcomp_i</i>		0.414*** (0.092)		0.422*** (0.113)
Observations (N)	6630	6630	5120	5120
Adj. <i>R</i> ²	0.511	0.522	0.498	0.514

Note: Cluster robust standard errors in parentheses. Includes time dummies, vendor fixed effects. Significance code: ****p* < 0.001, ***p* < 0.01, **p* < 0.05.

TABLE E2 | Robustness checks for quality spillover using propensity matching and synthetic control.

Dependent variable is <i>Defect_{it}</i>				
Variables	Matched suppliers		Synthetic Control	
Models	B	B4	B	B4
<i>Post_{it}</i>	0.311*** (0.074)	0.269*** (0.062)	0.377*** (0.067)	0.385*** (0.073)
PG1_{it}	−0.893*** (0.212)	−0.726*** (0.177)	−0.944*** (0.223)	−1.014*** (0.268)
PG1PG2_{it}	−0.211 (0.227)	−0.202 (0.193)	−0.251 (0.292)	−0.256 (0.311)
<i>Cumvol_{it}</i>	−0.127** (0.042)	−0.121** (0.039)	−0.206*** (0.033)	−0.217*** (0.036)
PG1PG2_{it} × supplier_locations_i		0.0963** (0.029)		0.101* (0.037)
PG1PG2_{it} × Product_commonality_i		0.512* (0.215)		0.614** (0.211)
PG1PG2_{it} × Process_commonality_i		0.604** (0.202)		0.677** (0.226)
Observations (N)	6630	6630	5120	5120
Adj. <i>R</i> ²	0.513	0.526	0.503	0.527

Note: Cluster robust standard errors in parentheses. Includes time dummies, vendor fixed effects. Significance code: ****p* < 0.001, ***p* < 0.01, **p* < 0.05.

TABLE E3 | Random effects DID estimate of quality improvement after SIU creation.

Variables	Dependent variable is $Defect_{it}$			
	Model A	Model A1	Model B (Truck)	Model B1 (Car)
	All suppliers to Car and Truck	Common suppliers to Car and Truck	Common & not-common suppliers to Truck	Common & not-common suppliers to Car
Post	0.832*** (0.139)	0.852*** (0.211)	0.906*** (0.154)	−0.752** (0.213)
Treat	0.708*** (0.104)	0.825*** (0.148)		
PT	−1.374*** (0.071)	−1.403*** (0.103)		
Supnum	−0.265*** (0.036)	−0.140*** (0.054)	−0.451*** (0.0450)	−0.0762 (0.0621)
Techcomp	−0.281*** (0.056)	−0.290** (0.069)	−0.173* (0.0726)	−0.344*** (0.086)
Cumvol	−0.411*** (0.030)	−0.401*** (0.0422)	−0.383*** (0.0382)	−0.396*** (0.0480)
Common			−0.0308 (0.119)	0.555* (0.248)
PC			0.00353 (0.0805)	−0.088 (0.188)
Time Dummies	yes	yes	yes	yes
Constant	12.33***	11.77***	8.564***	11.89***
Overall R^2	0.251	0.200	0.327	0.200

Note: Cluster robust standard errors in parentheses, *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

improvement levels of incoming quality of components between the estimated groups of suppliers over time after the organizational change has taken place. For the third estimation, the DID coefficient PC is not significant, indicating that there is no significant difference in the improvement levels of incoming quality of components between the common and not-common suppliers of the truck division. Put alternatively, the new structure of the car division has no significant effect on the improvement in quality supplied to the truck division—*there is no significant spillover*.

For the fourth estimation, the DID coefficient PC is again not significant, indicating that there is no significant difference in the improvement levels of incoming quality of components between the common and not-common suppliers of the car division. This shows that there is no significant difference between the quality improvement of a car supplier who supplies to truck, and who does not. Supplying to another buyer is immaterial, and quality improvement is sticky to a focal buyer–supplier dyad. These results provide strong support for our hypotheses and are completely aligned with fixed effect estimations.

Appendix F
Structural Elements of an Organic Versus a Mechanistic Structure
 See Table F1.

TABLE F1 | Structural elements of an organic versus a mechanistic structure and specific policy changes in the SIU.

Dimensions of organization structure	Mechanistic structure	Deployment of sourcing and supplier quality in truck	Deployment of SIU in car	Organic structure
Specialization of activities and tasks	<i>Activities are performed by someone with that function and no other.</i>	Quality Engineers perform incoming and final inspection. Materials Engineers focus on procurement logistics and component design support. Vendor Development engineers focus on new product development at suppliers.	SIU engineers required to perform general roles including quality audit at supplier and buyer, worker training. Kaizen deployment, process redesign including investment, and component design support.	<i>Generalists perform most functions, few specialists exist. Tasks adjusted and redefined to suit the environment.</i>
Standardization of processes	<i>Rigid, repetitive processes. Rewards dependent upon a fixed performance measurement system, which is reviewed at high frequency.</i>	Daily performance monitoring deployed to ensure that day-to-day production target, derived from monthly target, is met. Engineers work with tight roles with an objective to deliver the daily production target.	SIU Engineers have flexibility to deploy (a) process improvement, (b) quality assurance, and (c) design improvement initiatives to improve quality at assigned suppliers. Yearly review via quality improvement (reduction in defects) of assigned supplier group.	<i>Flexible processes; processes evolve as new situations unfold. Performance measurement is fluid.</i>
Formalization of decision making	<i>Formalized, written rules and standard operating procedures. Decision making draws from repetitive routines at lower levels.</i>	Formalized decision making, with daily decisions on sourcing taken at the operations head level, and deployed via department heads of Materials, Production, Quality, and Vendor Development to Engineers.	Engineers can deploy a range of changes at the vendor end to control defects and improve quality. Discussions on these improvements with SIU manager to be informal. Deployment of changes to be documented.	<i>Informal decision making. Subordinates decide on when and whether to convey decisions and information.</i>
Discretion (Centralization or Authority)	<i>Low discretion of subordinates, central authority makes decisions affecting the organization.</i>	Monthly production targets are reviewed daily for day-to-day performance by Production head with department heads. Deviations, if any, need approval of Production head, with subsequent confirmation from Plant manager.	Engineers can spend time on developing right solutions to problems at the vendors at their discretion. Experimentation with solutions is allowed. SIU manager to provide guidance, but confirmation from Plant manager for experiments or process changes not needed.	<i>Team members have high discretion in decision making. Lateral communication; consulting rather than command.</i>

Appendix G

Measures for Technical Complexity of Operations at Suppliers

This appendix provides an overview of our approach to measure the technical complexity of operations at suppliers. In the context of Auto Inc., we identify product complexity and manufacturing complexity as relevant factors that affect complexity at supplier operations. In line with Novak and Eppinger (2001), we measure product complexity on a spectrum of 0 to 1 (low product complexity to high product complexity) as the unweighted average of three characteristics: (1) the number of subcomponents, (2) product interactions, and (3) product novelty, each of which is scaled to take values from 0 to 1. The number of subcomponents in a product is a measure of the number of distinct elements (or sub-part) for each product sourced from a supplier. For each product supplied by a supplier, the number of subcomponents was identified. If the number of subcomponents was less than three, then the product received a score of 0; if the number of subcomponents was less than six and more than two, then the product received a score of 1; if the number of subcomponents was less than ten and more than five, then the product received a score of 2; and if the product had more than ten subcomponents, it received a score of 4. These scores were then scaled to 0, 1/3, 2/3, and 1 to represent the measure of the number of subcomponents for a product in our data set. Product interactions indicate the level of the product, whether it is a nuclear product, subassembly for a module, or a module. Products were scored 0 for nuclear products, 1 for subassembly of a module, and 2 for a module. The scores were then scaled to 0, 1/2, and 1 to represent the measure of product interactions. Product novelty denotes the extent to which the product involves technologies that are new to the involved supplier. Products were scored 0 if the supplier had prior experience with similar technologies, 1 if the supplier had some experience with similar technologies, and 2 if the supplier had no prior experience with the relevant technologies. The scores were then scaled to 0, 1/2, and 1 to indicate the measure of product novelty. Based on these three scores, the product complexity for a product “*i*” was computed as:

$$product_complexity_i = \frac{number_of_elements_of_a_component_i + product_interactions_i + productnovelty_i}{3}$$

Our discussions with the Car SIU team identified two critical factors that could increase the complexity of operations at suppliers. The first factor relates to the tolerance limits of the product, which gauges the extent of precision needed in relation to the functional dimensions of the product. Products that require tighter tolerances involve more precise

manufacturing, which in turn increases the complexity involved in operations. Products were scored 0 if critical dimensions of a component had to be manufactured to a precision within a millimeter or more, 1 if the manufacturing precision required was less than a millimeter by more than one-tenth of a millimeter, 2 if the manufacturing precision required was less than one-tenth of a millimeter and more than 10 μ m, and 3 if the manufacturing precision required was less than 10 μ m. The scores were then scaled to 0, 1/3, 2/3, and 1 to indicate the measure of tolerance limits. The second factor relates to the manufacturing sophistication required at supplier facilities to manufacture the product. If the manufacturing process requires unique tools, exclusive fixtures, and special processes to manufacture a product, it was scored 3; if only two of these features were required, it was scored 2; if only one of these features was required, it was scored 1; and if none of these features were required, it was scored 0. The scores were then scaled to 1, 2/3, 1/3, and 0 to indicate the measure of manufacturing sophistication. Based on these two scores, the manufacturing complexity for a product “*i*” was computed as:

$$manufacturing_complexity_i = \frac{tolerance_limits_i + manufacturing_sophistication_i}{2}$$

Finally, based on the number of distinct products “*n*” sourced from a supplier, we computed the technical complexity of supplier operations as:

$$Technical_Complexity_of_Supplier_Operations_i = \frac{1}{n} \sum_{i=1}^n \frac{product_complexity_i + manufacturing_complexity_i}{2}$$

Note that complexity can take values between 0 and 1. In our data, the technical complexity of supplier operations ranged from 0.2 (low complexity) to 1 (high complexity), with a mean of 0.69 and a standard deviation of 0.18.

Appendix H

Results for Regression Discontinuity Design (RDD) Analysis

See Figure H1.

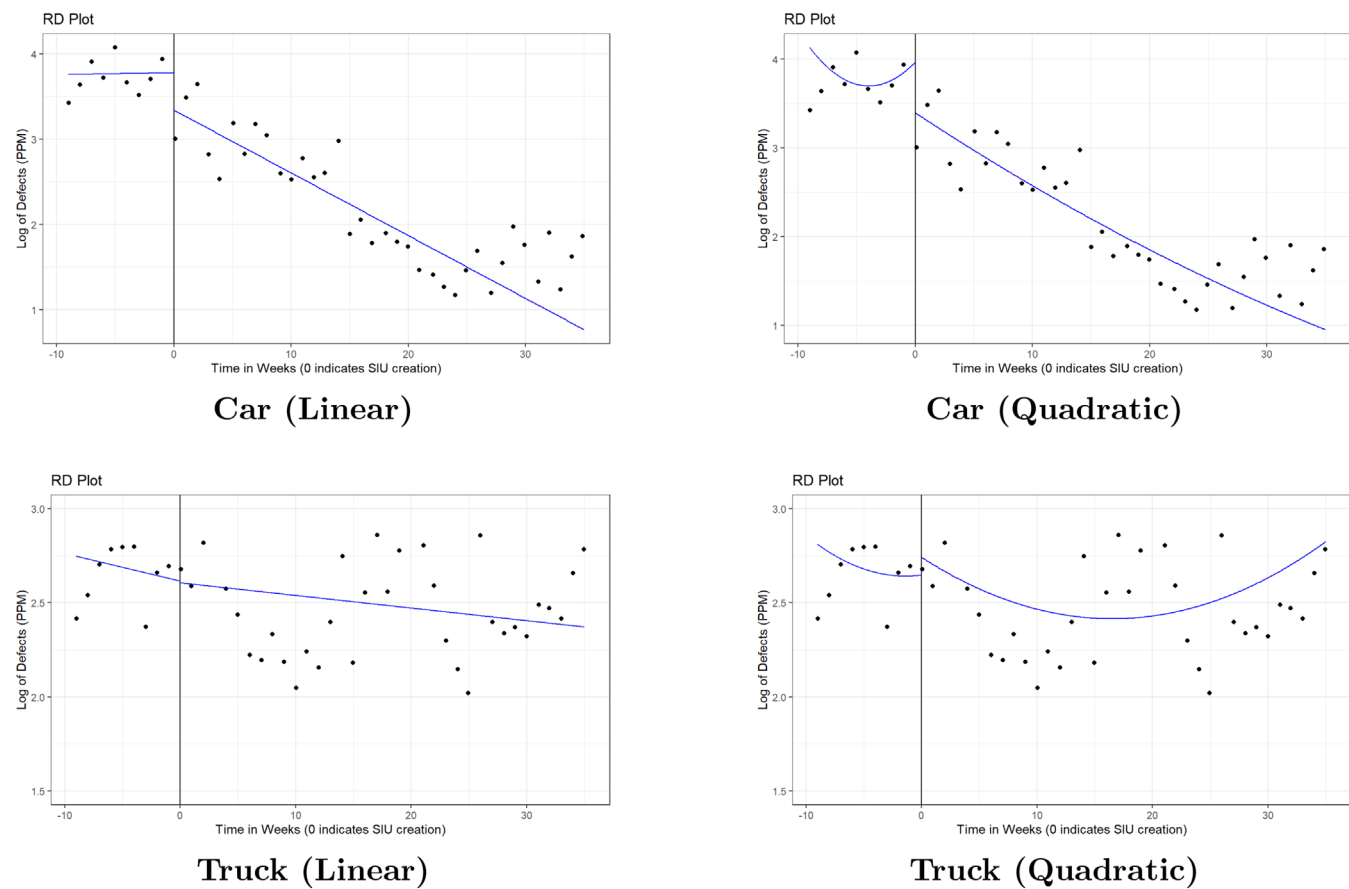


FIGURE H1 | Graphical RDD estimates for log of defects (first and second order local polynomial estimates).

TABLE H1 | RDD estimates for first order local polynomial regression.

Effect	Estimate	Std. error	Significance
Car			
Intercept	3.8468	0.1471	***
Discontinuity (D)	−0.6864	0.1608	***
Time (Pre-SIU)	0.0232	0.0262	
Time (Post-SIU)	−0.0815	0.0263	**
F-Statistic	263.5		***
Truck			
Intercept	2.7089	0.1041	***
Discontinuity (D)	−0.1549	0.1137	
Time (Pre-SIU)	0.0135	0.0185	
Time (Post-SIU)	−0.0157	0.0186	
F-Statistic	24.34		***

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.